

The Eye

Robert Folberg and Amy Y. Lin

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Although this chapter comes at the end of the book, it is not an afterthought. Vision is a major quality-of-life issue. Before the public awareness of acquired immunodeficiency syndrome (AIDS) and Alzheimer disease, the second most feared disease among Americans after cancer was blindness.

In general, diseases that produce vision loss do not attract as much attention as many of the life-threatening conditions described in this book. For example, age-related macular degeneration (AMD) is the most common cause of irreversible visual loss in the United States. Most individuals with AMD do not have a total loss of vision (i.e., an immersion into darkness). The histopathology is unimpressive: small scars develop in the macula. But consider the effect of these tiny macular scars for a retired schoolteacher with AMD. The central portion of their vision is lost. The faces of spouses or grandchildren are not visible. They cannot read a book or newspaper. Once a model of independence,

this teacher can no longer drive a car and must be chauffeured everywhere. In short, this person is robbed of the everyday joys many of us take for granted.

To study the eye, one needs to comprehend all that has come before. For example, the pathology of the eyelids builds on knowledge of dermatopathology (Chapter 25), and the pathology of the retina and optic nerve extends what was learned in Chapter 28 about the brain and central nervous system. However, the study of ocular pathology does more than merely repeat what has been presented thus far. **The eye provides the only site in which a physician can directly visualize a variety of microcirculatory disturbances ranging from arteriosclerosis to angiogenesis in the clinic.** Although some conditions are unique to the eye (e.g., cataracts and glaucoma), many ocular conditions share similarities with disease processes elsewhere in the body that are modified by the unique structure and function of the eye (Fig. 29.1).

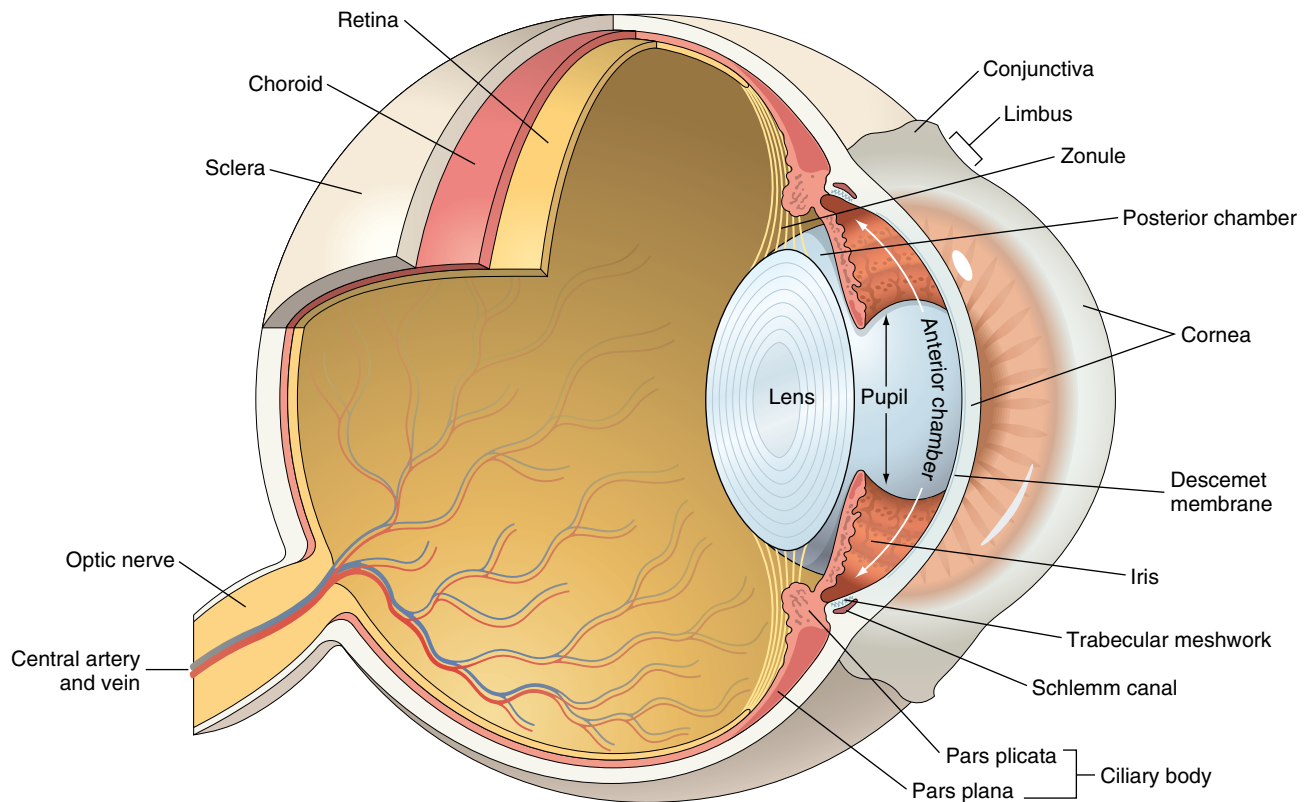


Fig. 29.1 Anatomy of the eye.

The complex composition of the eye (Fig. 29.1) and surrounding tissues presents a formidable challenge in teaching this material. On the one hand, one might take a conceptual approach based on the pathogenesis of specific disorders across the spectrum of ocular tissue types. For example, vitamin A deficiency is a leading cause of blindness in low-resource countries (Chapter 9). This condition affects the conjunctiva leading to severe dry eye (*xerophthalmia*), corneal clouding (*keratomalacia*), and retinal dysfunction leading to night blindness. In congenital rubella (Chapter 10), one pathogen may produce cataract, glaucoma, and retinal damage. Entire books are written about systemic diseases and the eye.

On the other hand, in many, if not most clinical settings, attention is centered on specific tissues in and around the eye. This chapter's organization by anatomic tissue type allows a focus on the pathologic basis of disease, as seen most commonly in clinical settings.

ORBIT

Proptosis

Diseases that increase orbital contents displace the eye forward, a condition known clinically as **proptosis**. Aside from the obvious cosmetic concerns, the proptotic eye might not be entirely covered by the eyelids, and the tear film might not be distributed evenly across the cornea. Chronic corneal exposure to air is injurious, leading to dryness and pain and predisposing to corneal ulceration and infection. Proptosis may be axial (displaces the eye directly forward) or non-axial (positional). For example, any enlargement of the lacrimal gland from inflammation (e.g., sarcoidosis) or neoplasm (e.g., lymphoma, pleomorphic adenoma, or adenoid cystic carcinoma) produces a proptosis that displaces the eye inferiorly and medially because the lacrimal gland is positioned superotemporally within the orbit.

Masses within the cone formed by the horizontal rectus muscles generate axial proptosis: the eye bulges straight forward. The two most common primary tumors of the optic nerve (a tract of the central nervous system), *glioma* and *meningioma*, produce axial proptosis because the optic nerve is positioned within the muscle cone. The orbital contents are subject to the same disease processes that affect other tissues. Representative inflammatory conditions and neoplasms of the orbit are discussed briefly next.

Thyroid Associated Orbitopathy (Graves Associated Orbitopathy)

Axial proptosis is an important clinical manifestation of Graves disease (Chapter 24). It is caused by the accumulation of extracellular matrix proteins and variable degrees of fibrosis in the rectus muscles (Fig. 29.2). The development of thyroid ophthalmopathy may sometimes be independent of thyroid function status.

Other Orbital Inflammatory Conditions

The floor of the orbit is the roof of the maxillary sinus, and the medial wall of the orbit—the lamina papyracea—separates the orbit from the ethmoidal sinuses. As a result, an uncontrolled sinus bacterial or fungal infection may spread to the orbit. This occurs most commonly in immunosuppressed individuals, and patients with diabetes, but may also occur rarely in persons without any predisposition. Systemic conditions such as *granulomatosis with polyangiitis* (Chapter 11) may present first in the orbit and may be confined there for prolonged periods; alternatively, it may involve the orbit secondarily by extension from the sinuses.

Idiopathic orbital inflammation (also known as orbital inflammatory pseudotumor, or simply orbital pseudotumor), is another inflammatory condition affecting the orbit without any identifiable local or systemic causes. This condition may be unilateral or bilateral and may

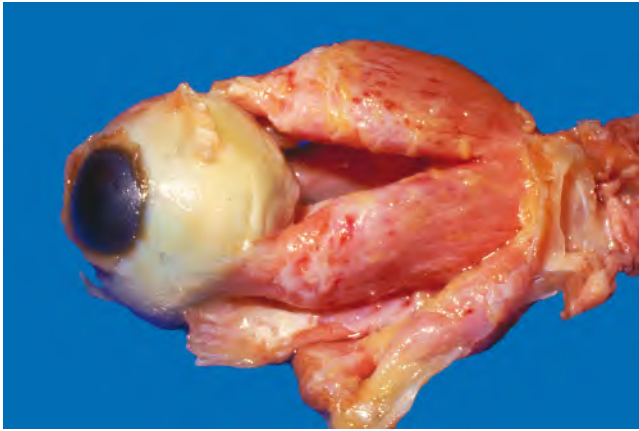


Fig. 29.2 Thyroid eye disease. The extraocular muscles are greatly distended in this postmortem dissection of tissues from a patient with thyroid (Graves) ophthalmopathy. Note that the tendons of the muscles are spared. (Courtesy Dr. Ralph C. Eagle Jr, Wills Eye Hospital, Philadelphia.)

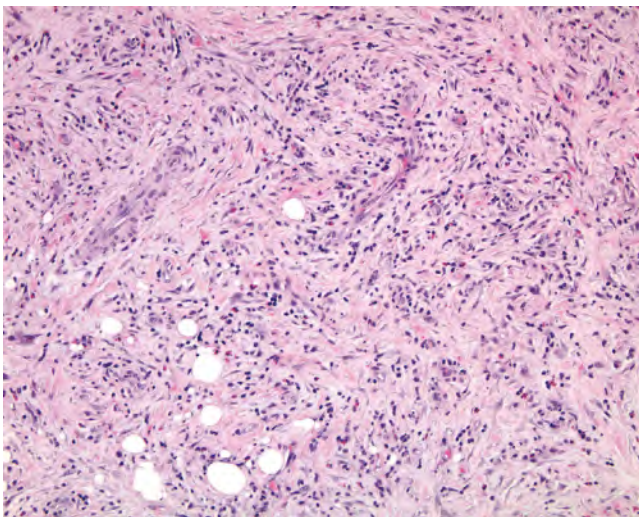


Fig. 29.3 Idiopathic orbital inflammation (orbital inflammatory pseudotumor). The orbital fat is replaced by fibrosis. Note the chronic inflammation, accompanied in this case by eosinophils.

affect all orbital tissue elements or may be confined to the lacrimal gland (*sclerosing dacryoadenitis*), extraocular muscles (*orbital myositis*), or Tenon capsule, the fascial layer that wraps around the eye (*posterior scleritis*). IgG4-related disease (Chapter 6) should be excluded before declaring an orbital inflammation idiopathic.

MORPHOLOGY

Idiopathic orbital inflammation is characterized histologically by chronic inflammation and variable degrees of fibrosis (Fig. 29.3). The inflammatory infiltrate typically includes lymphocytes and plasma cells, and occasionally eosinophils. Germinal centers, when present, suggest reactive lymphoid hyperplasia. Vasculitis may be present, raising the possibility of an underlying systemic condition. The presence of necrosis, degenerating collagen, and vasculitis should raise the suspicion of granulomatosis with polyangiitis. Idiopathic orbital inflammation is typically confined to the orbit but may develop concomitantly with sclerosing inflammation in the retroperitoneum, the mediastinum, and the thyroid, especially as a manifestation of IgG4-related disease.

Neoplasms

The most frequently encountered primary neoplasms of the orbit are vascular in origin: capillary hemangioma of infancy and early childhood and lymphangioma (both unencapsulated) and encapsulated cavernous hemangioma found typically in adults. These are described in other chapters. Only a handful of orbital masses are encapsulated (e.g., pleomorphic adenoma of the lacrimal gland, dermoid cyst, neurilemmoma), and the recognition of encapsulation on imaging studies allows the surgeon to anticipate pathologic findings.

Non-Hodgkin lymphoma, like idiopathic orbital inflammation, can affect the entire orbit or be confined to the orbit's compartments, such as the lacrimal gland. Orbital lymphomas are classified according to the World Health Organization (WHO) classification system (Chapter 13).

Primary orbital malignancies may arise in orbital tissues and are classified according to the scheme used for the parent tissue. For example, the lacrimal gland may be considered a minor salivary gland, and lacrimal gland tumors are classified as salivary gland tumors.

Metastases to the orbit may present distinctive signs and symptoms that point to the origin of the tumor. For example, metastatic prostatic carcinoma may present clinically like idiopathic orbital inflammation, while metastatic neuroblastoma and Wilms tumor (richly vascularized neoplasms) may produce characteristic periorbital ecchymoses. Sinus neoplasms may also invade the orbit.



KEY CONCEPTS

Proptosis, Inflammation, and Primary Tumors of the Orbit

- Proptosis results from lesions or pathologic changes in tissue that occupy space in the orbit.
- Inflammation in the orbit may develop by extension of local disease in adjacent tissues (e.g., sinusitis) or as a component of systemic disease (e.g., granulomatosis with polyangiitis).
- The most common primary tumors of the orbit are vascular (e.g., capillary and cavernous hemangiomas).

EYELID

The eyelid is composed of skin externally and mucosa (the conjunctiva) on its underlying surface (Fig. 29.4). In addition to covering and protecting the eye, elements within the eyelid generate critical tear film components. For example, the sebaceous glands in the eyelid produce lipid that slows the evaporation of tears. The drainage system of the sebaceous (*Meibomian*) glands may be obstructed by chronic inflammation at the eyelid margin (*blepharitis*), leading to atrophy of these glands. A reduced surface tear lipid layer may lead to dry eye because of accelerated evaporation.

When the Meibomian gland ductules are obstructed at the eyelid margin by blepharitis or, less commonly, by neoplasm, lipid may extravasate into surrounding tissue and provoke a granulomatous response producing a lipogranuloma or *chalazion* (a small and firm painless nodule on the eyelid).

Neoplasms

The most common malignancy of the eyelid is basal cell carcinoma (see Fig. 25.15). Surprisingly, primary melanomas of the eyelid skin are extremely rare. Regardless of histogenesis, eyelid neoplasms may distort tissue and prevent the eyelids from closing completely. Because chronic exposure when the eyelids do not close properly damages the cornea, prompt treatment of locally invasive basal cell carcinomas is imperative to preserve vision. Basal cell carcinoma has a distinct predilection for the lower eyelid and the medial canthus.

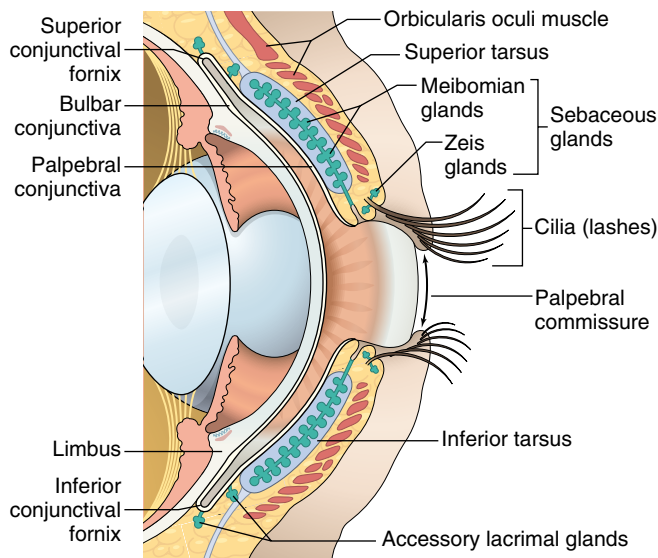


Fig. 29.4 Anatomy of the conjunctiva and eyelids.

Sebaceous carcinoma may form a local mass mimicking chalazion (a localized swelling caused by obstruction of sebaceous glands) or diffusely thicken the eyelid. This neoplasm may also resemble inflammatory processes such as blepharitis or ocular cicatricial pemphigoid because of a predilection for intraepithelial spread as occurs in Paget disease of the nipple (Chapter 23) or vulva (Chapter 22). Sebaceous carcinoma tends to spread first to the parotid and submandibular nodes. The overall mortality rate can be as high as 22%. Sebaceous carcinoma of the eyelid is less likely to be associated with Muir-Torre syndrome, a rare inherited cancer-prone syndrome caused by germline mutations in mismatch repair genes, than sebaceous neoplasms developing elsewhere in the skin.

MORPHOLOGY

In moderately differentiated or well-differentiated sebaceous carcinoma, cytoplasmic vacuolization, when present, is a helpful clue. This cancer may, however, resemble a variety of other malignancies histologically, including basal cell carcinoma; hence establishing the correct diagnosis can be difficult. **Pagetoid spread** (Fig. 29.5) may mimic Bowenoid actinic keratosis in the eyelid and carcinoma in situ in the conjunctiva. Sebaceous carcinoma may spread through the conjunctival epithelium and the epidermis to the lacrimal drainage system and the nasopharynx. It may also extend into the lacrimal gland ductules and the main lacrimal gland.

In individuals with AIDS, *Kaposi sarcoma* may develop in either the eyelid or the conjunctiva. In the eyelid, the lesion may appear purple because the vascular lesion is embedded in the dermis whereas in the thin mucous membrane of the conjunctiva, Kaposi sarcoma appears bright red and may be confused clinically with a subconjunctival hemorrhage.

KEY CONCEPTS

Carcinoma of the Eyelid

- Basal cell carcinoma is the most common primary malignancy of the eyelid and may be locally invasive but rarely metastasizes.
- By contrast, sebaceous carcinoma of the eyelid may metastasize and is, therefore, a serious and potentially life-threatening condition.

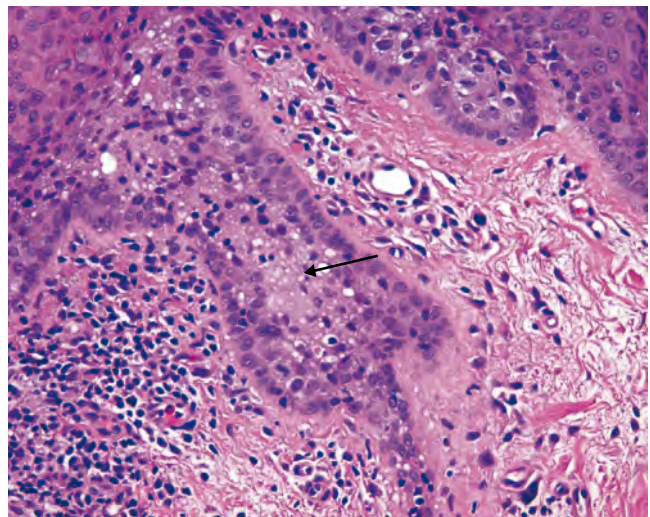


Fig. 29.5 Pagetoid spread of sebaceous carcinoma. Neoplastic cells with foamy cytoplasm are present within the epidermis (arrow). Invasive sebaceous carcinoma was identified elsewhere in this biopsy sample.

CONJUNCTIVA

The conjunctiva is divided into zones (Fig. 29.4), each with distinctive histologic features and responses to injury. The conjunctiva lining the interior of the eyelid, the *palpebral conjunctiva*, is normally covered by a low-profile pseudostratified epithelium, which is tightly tethered to the tarsus. The palpebral conjunctiva may respond to inflammation, such as in allergic or bacterial conjunctivitis, by being thrown into minute papillary folds. The conjunctiva in the *fornix* is a pseudostratified columnar epithelium rich in goblet cells. The fornix also contains accessory lacrimal tissue, and the ductules of the main lacrimal gland empty through the conjunctiva in the fornix superiorly and laterally. The lymphoid population of the conjunctiva is most noticeable in the fornix, and in *viral conjunctivitis*, lymphoid follicles may enlarge sufficiently to be visualized clinically by slit-lamp examination. Primary lymphoma of the conjunctiva (typically indolent marginal zone B-cell lymphoma) is most likely to develop in the fornix. The *bulbar conjunctiva*—the conjunctiva that covers the eye's surface—is a nonkeratinizing stratified squamous epithelium. The limbus, the intersection between the sclera and cornea, also marks the transition between conjunctival and corneal epithelium (Fig. 29.1). Ocular surface stem cells reside at the limbus, which is commonly the site of ocular surface squamous cell neoplasia.

The conjunctiva, like the eyelid, is richly invested with lymphatic channels. Malignant neoplasms in the eyelid and conjunctiva tend to spread to regional lymph nodes (parotid and submandibular node groups).

Conjunctivitis-Trachoma and Other Causes of Painful Blindness

Trachoma is caused by obligate intracellular bacterium, Chlamydia trachomatis. It is a leading cause of preventable blindness worldwide. Infection spreads through personal contact (via hands, clothes, bedding or hard surfaces) and by flies that have been in contact with discharge from the eyes or nose of an infected person. Untreated trachoma and associated conjunctivitis may lead to significant scarring. If severe, the scarring may obliterate both the fornix (resulting in loss of goblet cells and mucin production) and the lacrimal ductules (diminishing tear production). Eyelashes may also be misdirected to the cornea (*trichiasis*). The result is an end-stage severely dry and chronically irritated eye, which leads to painful blindness.

Conjunctival scarring is also seen after exposure of the ocular surface to caustic alkalis or as a sequela to ocular cicatricial *pemphigoid* (Chapter 25).

The conjunctiva may be scarred iatrogenically through reaction to drugs or as a consequence of surgery. Cancer surgery in other parts of the body requires excision of the lesion with a margin of normal tissue to ensure complete removal. However, extensive surgical excision of conjunctival tumors can remove many goblet cells or compromise lacrimal gland ductules that traverse the conjunctiva. Thus, removing a conjunctival neoplasm or a precursor lesion may leave the affected individual with a painful dry eye that can compromise vision. Therefore, surgeons often remove only the invasive components of conjunctival neoplasms and treat intraepithelial involvement with tissue-sparing modalities such as cryotherapy or topical chemotherapy delivered as eye drops.

Pterygium and Pinguecula

Both *pterygium* and *pinguecula* appear as submucosal elevations on the conjunctiva resulting from actinic damage and therefore located in sun-exposed regions of the conjunctiva (i.e., in the fissure between both the upper and the lower eyelids—the interpalpebral fissure). Pinguecula is a small, yellowish submucosal elevation and appears astride the limbus (i.e., the growth next to, but not overlapping, the cornea). Pterygium typically originates in the conjunctiva astride the limbus. It is a wedge-shaped submucosal growth of *fibrovascular connective tissue that migrates onto the cornea*, dissecting into the plane occupied normally by the Bowman layer. Pterygium does not cross the pupillary axis and, aside from the possible induction of mild astigmatism, does

not threaten vision. Although most pterygia are entirely benign, they can cause redness and irritation of the eye. Submitting the excised tissue for pathologic examination is worthwhile because, occasionally, precursors of actinic-induced neoplasms (i.e., squamous cell carcinoma and melanoma) are detected in these lesions.

Neoplasms

Both squamous and melanocytic neoplasms and their precursors tend to develop at the limbus. Conjunctival *squamous cell carcinoma* may be preceded by intraepithelial neoplastic changes analogous to those seen in the evolution of cervical squamous cell carcinoma. In the conjunctiva, the spectrum of changes from mild dysplasia to carcinoma in situ is called *ocular surface squamous neoplasia* (OSSN). Squamous papillomas of the conjunctiva have the same morphology as mucosal papillomas elsewhere in the body. Squamous papillomas and conjunctival intraepithelial neoplasia may be associated with the presence of high-risk human papillomavirus (HPV) types 16 and 18.

Conjunctival nevi are encountered commonly but seldom invade the cornea or appear in the fornix or over the palpebral conjunctiva. Pigmented lesions in these zones of the conjunctiva most likely represent melanomas or melanoma precursors. Conjunctival compound nevi characteristically contain subepithelial cysts lined by surface epithelium (Fig. 29.6A,B). In late childhood or adolescence, conjunctival nevi may acquire an inflammatory component rich in lymphocytes, plasma cells, and eosinophils. The resultant *inflamed juvenile nevus* is entirely benign.

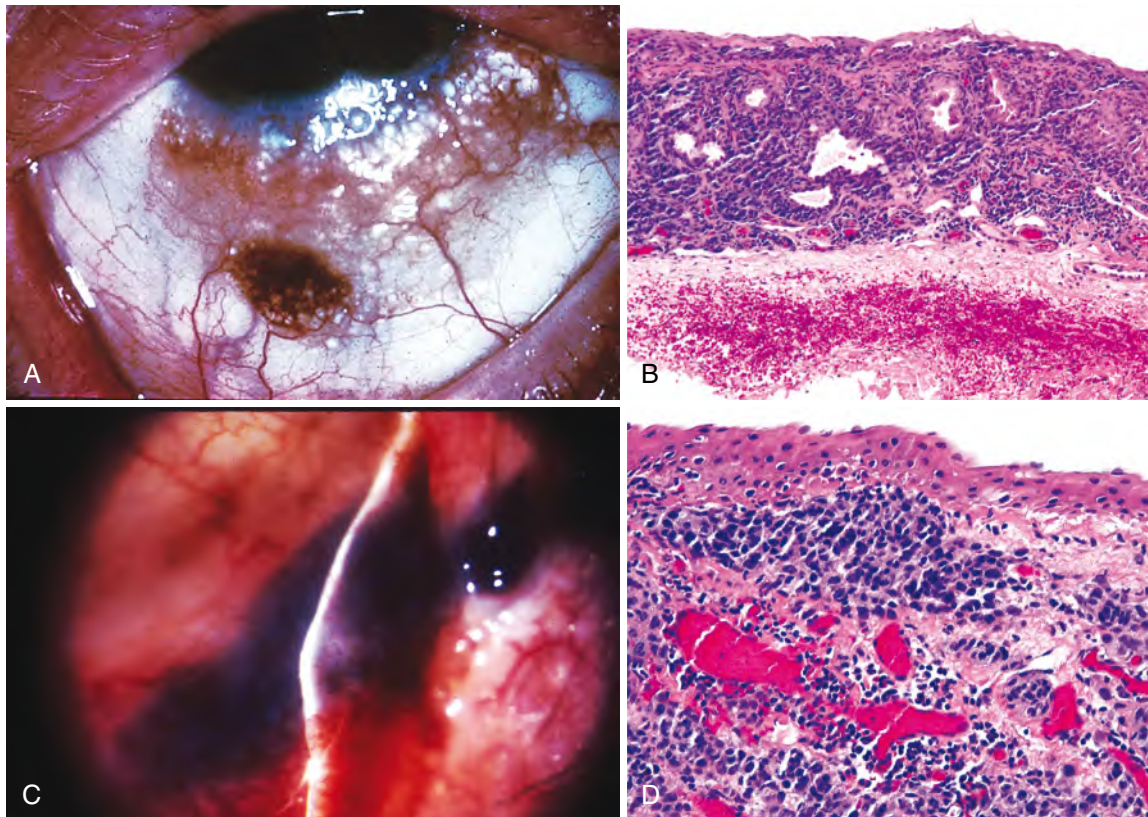


Fig. 29.6 Melanocytic lesions involving conjunctiva. (A–B) Cystic compound nevus of the conjunctiva. (C–D) Conjunctival malignant melanoma. In C, note the deflection of the beam of the slit lamp over the surface of the lesion, indicative of invasion. (A–B, From Folberg R, Jakobiec FA, Bernardino VB, Iwamoto T. Benign conjunctival melanocytic lesions: clinicopathologic features. *Ophthalmology*. 1989;96:436-461.)

Conjunctival melanomas are unilateral neoplasms, typically affecting fair-complexioned individuals in middle age (Fig. 29.6C,D). Most cases develop through a phase of intraepithelial growth termed *conjunctival melanocytic intraepithelial neoplasia (C-MIN)*. Between 50% and 90% of individuals with incompletely treated primary acquired melanosis with atypia develop conjunctival melanoma. *BRAF* V600E mutations are present in nearly 40% of conjunctival melanomas, a feature shared with cutaneous melanoma (Chapter 25). The lesions tend to spread first to the parotid or submandibular lymph nodes. Approximately 25% of conjunctival melanomas prove to be fatal.

KEY CONCEPTS

Conjunctiva

- Conjunctival scarring, a consequence of various conditions, may result in painful loss of vision by interfering with the delivery and maintenance of the tear film.
- Many conjunctival neoplasms originate at the limbus, which contains conjunctival stem cells.
- Conjunctival malignancies, especially conjunctival melanomas, tend to spread through the rich lymphatics of the conjunctiva to regional lymph nodes.

SCLERA

The sclera consists mainly of collagen and contains few blood vessels and fibroblasts; hence wounds and surgical incisions tend to heal poorly. Immune complex deposits within the sclera, such as in rheumatoid arthritis, may produce necrotizing *scleritis*.

The sclera may appear “blue” in a variety of conditions. Some of these are:

- *Disorders that produce thinning of the sclera.* These include inflammation of the sclera (*scleritis*), exceptionally high intraocular pressure, and osteogenesis imperfecta (Chapter 26).
- *Heavily pigmented congenital nevus* of the underlying uvea, a condition known as *congenital melanosis oculi*. When accompanied by periocular cutaneous pigmentation, this condition is known as *nevus of Ota*.

CORNEA

The cornea and its overlying tear film—not the lens—make up the primary refractive surface of the eye (Fig. 29.7). Parenthetically, *myopia* typically develops because the eye is too long for its refractive power, and *hyperopia* results from an eye that is too short. The popularity of procedures such as laser-assisted in situ keratomileusis (LASIK) to sculpt the cornea and change its refractive properties attests to the importance of corneal shape in contributing to the eye’s refractive power.

Anteriorly, the cornea is covered by *epithelium* that rests on a basement membrane. The *Bowman layer*, situated just beneath the epithelial basement membrane, is acellular and forms an efficient barrier against the penetration of malignant cells from the epithelium into the underlying stroma.

The *corneal stroma* lacks blood vessels and lymphatics, contributing to the cornea’s transparency and the high rate of corneal transplantation success. Indeed, nonimmunologic graft failure (associated with loss of endothelial cells and subsequent corneal edema) is seen more commonly than immunologic graft rejection. The risk of corneal graft rejection increases with stromal vascularization and inflammation. A precise alignment of collagen in the corneal stroma also contributes to transparency.

Corneal vascularization may accompany chronic corneal edema, inflammation, and scarring. Scarring and edema disrupt the spatial

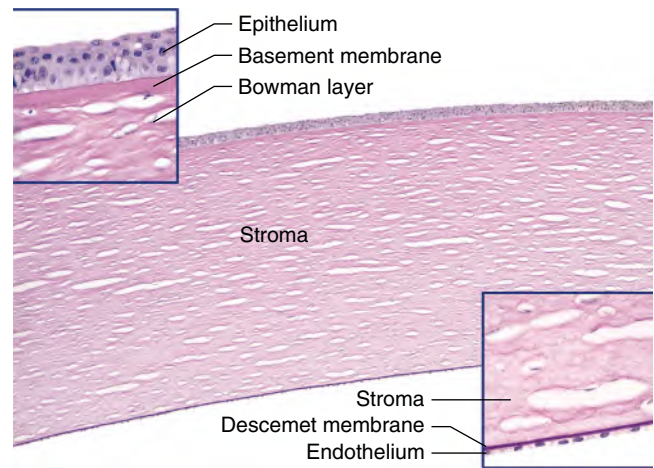


Fig. 29.7 Normal corneal microarchitecture. The corneal tissue is stained by periodic acid–Schiff (PAS) to highlight basement membranes. The inset at the upper left is a high magnification of the anterior layers of the cornea: epithelium, Bowman layer, and stroma. A very thin PAS-positive basement membrane separates the epithelium from the Bowman layer. Note that the Bowman layer is acellular. The inset at the lower right is a high magnification of the PAS-positive Descemet membrane and the corneal endothelium. The “holes” in the stroma are artifactitious spaces between parallel collagenous stromal lamellae.

alignment of stromal collagen and contribute to corneal opacification. Scars may result from trauma or inflammation. Usually, the corneal stroma is in a state of relative dehydration, mainly maintained by the active pumping of fluid from the stroma back into the anterior chamber by the corneal endothelium.

The corneal *endothelium* is derived from the neural crest and is unrelated to vascular endothelium. It rests on its basement membrane, referred to as *Descemet membrane*. A decrease in endothelial cells or a malfunction of endothelium results in stromal edema, which may be complicated by bullous separation of the epithelium (*bullous keratopathy*). Descemet membrane increases in thickness with age. It is the site of copper deposition in the Kayser-Fleischer ring of Wilson disease (Chapter 18).

Keratitis and Ulcers

Various pathogens (i.e., bacteria, fungi, viruses [especially herpes simplex and herpes zoster], and protozoa [*Acanthamoeba*]) can cause corneal ulceration. Acute keratitis may be accompanied by the dissolution of the corneal stroma, accelerated by the activation of collagenases within the corneal epithelium and stromal fibroblasts (also known as keratocytes). The specific forms of keratitis may have certain distinctive features. For instance, chronic herpes simplex keratitis may be associated with a granulomatous reaction involving the Descemet membrane (Fig. 29.8).

Corneal Degenerations and Dystrophies

Ophthalmologists have traditionally divided many corneal disorders into degenerations and dystrophies. Corneal degenerations may be either unilateral or bilateral and are typically nonfamilial. By contrast, corneal dystrophies are generally bilateral and are hereditary. Corneal dystrophies may affect selective corneal layers, or the changes may be distributed throughout multiple layers.

Band Keratopathies

Two types of band keratopathy serve as examples of corneal degenerations. *Calcific band keratopathy* is characterized by calcium deposition

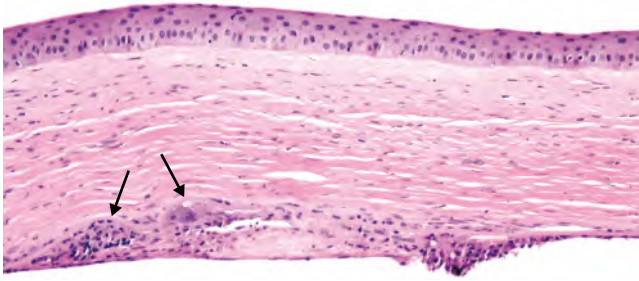


Fig. 29.8 Chronic herpes simplex keratitis. The cornea is thin and scarred (note the increased number of fibroblast nuclei). Granulomatous reaction in the Descemet membrane (arrows), is a histologic hallmark of chronic herpes simplex keratitis.

in the Bowman layer. This condition may complicate chronic uveitis and, less commonly, systemic hypercalcemia. *Actinic band keratopathy* develops in individuals exposed chronically to high levels of ultraviolet light. In this condition, extensive solar elastosis develops in the superficial layers of corneal collagen in the sun-exposed interpalpebral fissure; hence the horizontally distributed band of pathology. Similar to pinguecula, the sun-damaged collagen of the cornea may take on a yellow hue to the point that this condition is sometimes erroneously called “oil-droplet keratopathy.”

Keratoconus

With an incidence 1 in 2000 individuals, *keratoconus* is a fairly common disorder characterized by progressive thinning and ectasia of the cornea without evidence of inflammation or vascularization. Such thinning results in a cornea that has a conical rather than spherical shape. This abnormal shape generates irregular astigmatism that is difficult to correct with spectacles. Rigid contact lenses provide a smooth, spherical surface to the cornea and may provide refractive relief for individuals with keratoconus. Patients whose vision cannot be corrected with spectacles or contact lenses are candidates for corneal transplantation, which has a high degree of success in this condition. Unlike many types of degeneration, keratoconus is typically bilateral. It is associated with Down syndrome, Marfan syndrome, and atopic disorders. Its development may stem from a genetic predisposition superimposed by an environmental insult, such as eye rubbing in response to atopic conditions.

MORPHOLOGY

Thinning of the cornea with breaks in the Bowman layer is the histologic hallmark of keratoconus (Fig. 29.9). In some patients, the Descemet membrane may rupture precipitously, allowing the aqueous humor in the anterior chamber to gain access to the corneal stroma. The sudden effusion of aqueous humor through a gap in the Descemet membrane—**corneal hydrops**—may also cause vision to worsen suddenly. An episode of hydrops may be followed by corneal scarring that can also contribute to visual loss. Acute corneal hydrops can complicate Descemet membrane ruptures that develop secondary to extraordinary intraocular pressure elevations in **infantile glaucoma (Haab striae)** or following the now uncommon obstetric forceps injury to the eye.

Fuchs Endothelial Dystrophy

Fuchs endothelial dystrophy, one of several dystrophies, results from the loss of endothelial cells and the resulting edema and thickening of the stroma. It is one of the principal indications for corneal transplantation in the United States. The two primary clinical manifestations of Fuchs endothelial dystrophy, *stromal edema* and

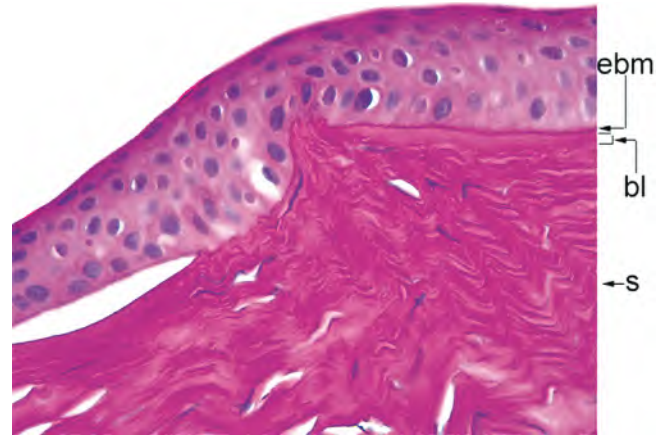


Fig. 29.9 Keratoconus. The tissue section is stained by periodic acid-Schiff to highlight the epithelial basement membrane (ebm), which is intact; the Bowman layer (bl), situated between the epithelial basement membrane; and the stroma (s). Following the Bowman layer from the right side of the photomicrograph toward the center, there is a discontinuity diagnostic of keratoconus. The epithelial separation just to the left of the Bowman layer discontinuity resulted from an episode of corneal hydrops caused by a break in the Descemet membrane (not shown).

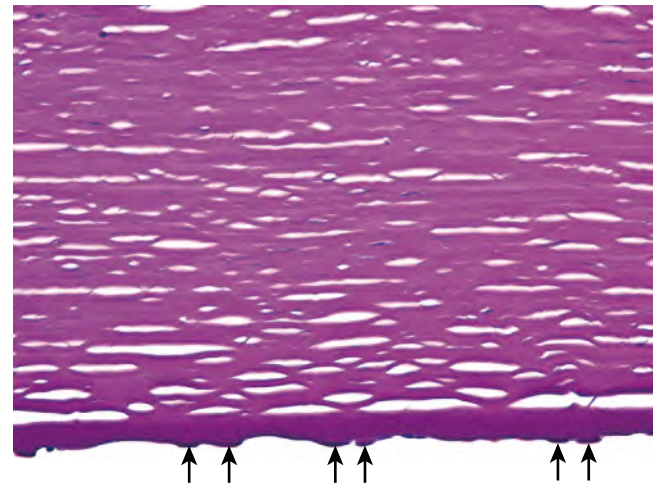


Fig. 29.10 Fuchs dystrophy. This tissue section is stained by periodic acid-Schiff to highlight the Descemet membrane, which is thick. Numerous drop-like excrescences (arrows), guttata, protrude downward from the Descemet membrane. Endothelial cell nuclei are not seen. Epithelial bullae, not shown in this micrograph, were present, reflecting corneal edema.

bullous keratopathy, are directly related to endothelial cell dysfunction and loss. Early in the disease, endothelial cells produce drop-like deposits of abnormal basement membrane material (*guttata*) that can be visualized clinically by slit-lamp examination. With disease progression, there is a decrease in the total number of endothelial cells, and the residual cells are incapable of maintaining stromal dehydration. Consequently, the stroma becomes edematous and thickens; it acquires a ground-glass appearance clinically, and vision is blurred (Fig. 29.10). Because of chronic edema, the stroma may eventually become vascularized. The number of endothelial cells may occasionally decrease following cataract surgery, even in individuals who do not have early forms of Fuchs dystrophy. This condition, *pseudophakic bullous keratopathy*, is also a common indication for corneal transplantation.

KEY CONCEPTS

Cornea

- The cornea—not the lens—is the primary refractive surface of the eye. Keratoconus is an example of a condition that distorts the cornea's contour and alters this refractive surface, producing an irregular form of astigmatism.
- The normal cornea is avascular, contributing to transparency and the low incidence of graft rejection after corneal transplantation.
- Corneal dystrophies are generally bilateral and inherited, while degenerations are typically acquired. Fuchs dystrophy and pseudophakic bullous keratopathy both produce visual loss through a common pathway of corneal edema, and both of these conditions are leading indications for corneal transplantation in the United States.

ANTERIOR SEGMENT

The anterior chamber is bounded anteriorly by the cornea, laterally by the trabecular meshwork, and posteriorly by the iris (Fig. 29.11). Aqueous humor, formed by the pars plicata of the ciliary body, enters the posterior chamber, bathes the lens, and circulates through the pupil to access the anterior chamber. The posterior chamber lies behind the iris and in front of the lens.

The lens is a closed epithelial system; the basement membrane of the lens epithelium (known as the lens capsule) envelops the lens. Thus, the lens epithelium does not exfoliate like the epidermis or mucosal epithelium. Instead, the lens epithelium and its derivative fibers accumulate within the confines of the lens capsule, thus “infolding.” With aging, therefore, the size of the lens increases. Neoplasms of the lens have not been described.

Cataract

The term *cataract* describes lenticular opacities that may be congenital or acquired. Systemic diseases (e.g., galactosemia, diabetes mellitus, Wilson disease, and atopic dermatitis), drugs (mainly corticosteroids), radiation, trauma, and many intraocular disorders are associated with cataracts. Age-related cataract typically results from opacification of the lens nucleus (*nuclear sclerosis*). The accumulation of urochrome pigment may render the lens nucleus brown, thus distorting the individual's perception of blue color (the predominance of

yellow hues in Rembrandt's paintings later in life might have been a consequence of nuclear sclerotic cataracts). Other physical changes in the lens may generate opacities. For example, the lens cortex may liquefy. Migration of the lens epithelium posterior to the lens equator may result in *posterior subcapsular cataract* secondary to enlargement of abnormally positioned lens epithelium. The technique most commonly used to remove opacified lenses extracts the lens contents, leaving the capsule intact. A prosthetic intraocular lens is typically inserted into the eye.

Anterior Segment Inflammation

By direct clinical observation, the eye allows us to see circulatory responses to acute and chronic inflammation (Fig 3.16). For example, acute corneal infections may produce an inflammatory exudate from the iris and ciliary body vessels. Leukocytes may settle out due to gravity and form a white layer in the lower portion of the anterior chamber that is visible by a penlight examination (*hypopyon*). Unlike the infected cornea, the hypopyon seldom contains organisms.

The slit lamp allows an optical view of the vascular response to chronic inflammation. Light reflects off inflammatory cells that appear to sparkle against a gray background. The aqueous humor is normally transparent, but exudate produces a murky appearance. Clinically, this finding by slit lamp examination is known as “cell and flare” and can be seen after blunt trauma to the eye, corneal inflammations, and inflammation of the iris or ciliary body. Inflammatory cells may sometimes adhere to the corneal endothelium, forming clinically visible *keratic precipitates*.

Just as pleural exudate in acute bronchopneumonia can lead to adhesions between the visceral and parietal pleura, the presence of exudate in the anterior chamber can lead to the formation of adhesions between the iris and the trabecular meshwork or cornea (*anterior synechiae*) or between the iris and the anterior surface of the lens (*posterior synechiae*). Anterior synechiae can lead to elevation in intraocular pressure sufficient to produce optic nerve damage. Prolonged contact between the iris and the anterior surface of the lens can deprive the lens epithelium of contact with aqueous humor. These *posterior synechiae* can induce fibrous metaplasia of the lens epithelium, which is clinically appreciated as *anterior subcapsular cataract* (Fig. 29.11). The pharmacologic induction of pupillary dilation and cycloplegia in

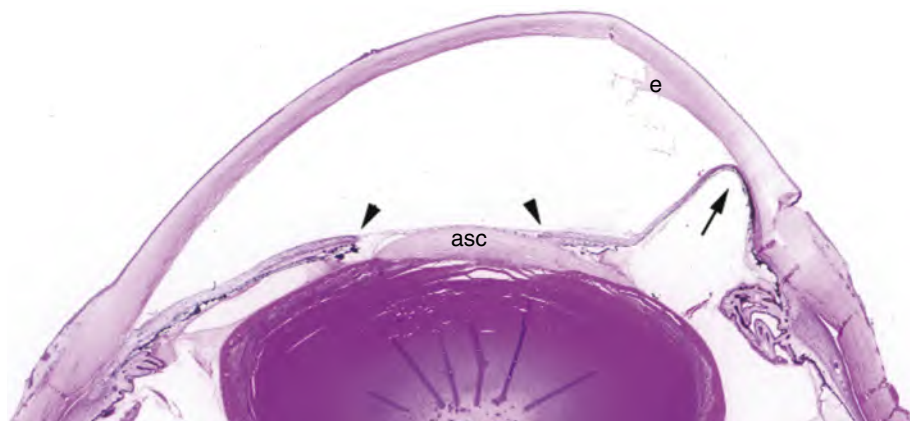


Fig. 29.11 Sequelae of anterior segment inflammation. This eye was removed for complications of chronic corneal inflammation (not visible at this magnification). The exudate (*e*) in the anterior chamber would have been visualized with a slit lamp as an optical “flare.” The iris is adherent focally to the cornea, obstructing the trabecular meshwork (anterior synechia, *arrow*) and to the lens (posterior synechiae, *arrowheads*). An anterior subcapsular cataract (*asc*) has formed. The radial folds in the lens are artifacts.

individuals with anterior segment inflammation prevents synechiae formation and their sequelae.

Although inflammation confined to the anterior segment is technically intraocular, the diagnosis of *endophthalmitis* is reserved for inflammation involving the vitreous humor.

Glaucoma

Glaucoma is a name for disease processes characterized by distinctive changes in the visual field and the optic nerve cup that are usually associated with elevated intraocular pressure. However, some individuals with normal intraocular pressure may develop the same characteristic optic nerve and visual field changes (*normal or low-tension glaucoma*). The relationship between intraocular pressure and optic nerve damage is discussed later under Optic Nerve.

Pathogenesis

To understand the pathophysiology of glaucoma, it is helpful to consider the formation and drainage of aqueous humor. As Fig. 29.12 illustrates, aqueous humor is produced in the ciliary body and passes from the posterior chamber through the pupil into the anterior chamber. Although there are multiple pathways for the egress of fluid from the anterior chamber, most of the aqueous humor drains through the trabecular meshwork situated in the angle formed by the intersection between the corneal periphery and the anterior surface of the iris. With this background, glaucoma can be classified into two major categories.

- The aqueous humor has complete physical access to the trabecular meshwork in *open-angle glaucoma*; the elevation in intraocular pressure results from increased resistance to aqueous outflow

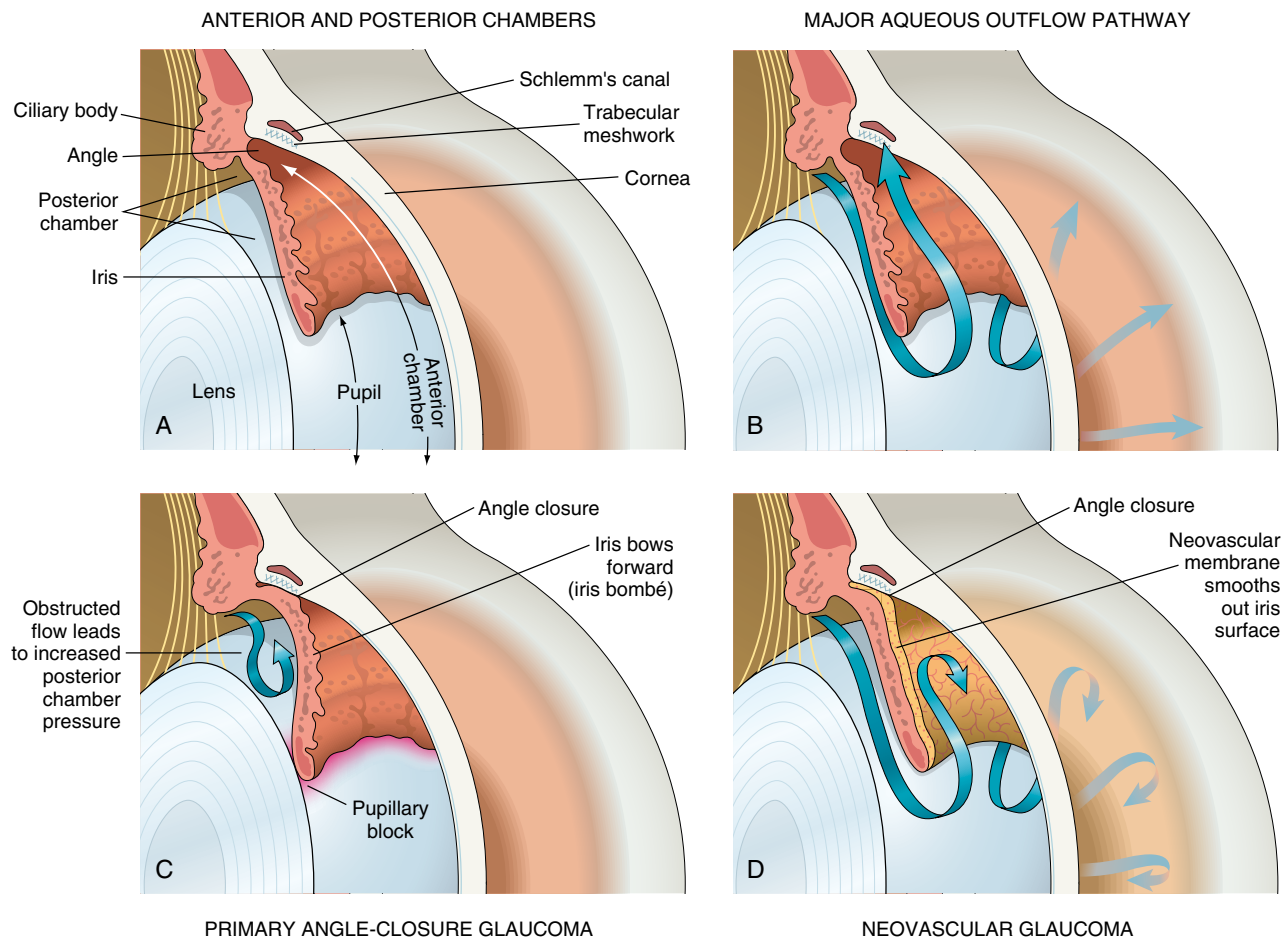


Fig. 29.12 Pathogenesis of glaucoma. (A) The normal eye. Note that the surface of the iris is highly textured with crypts and folds. *Upper right*, The normal flow of aqueous humor. (B) Aqueous humor, produced in the posterior chamber, flows through the pupil into the anterior chamber. The major pathway for the egress of aqueous humor is through the trabecular meshwork, into the Schlemm canal. Minor outflow pathways (uveoscleral and iris, not depicted) contribute to a limited extent to aqueous outflow. (C) Primary angle-closure glaucoma. In anatomically predisposed eyes, transient apposition of the iris at the pupillary margin to the lens blocks the passage of aqueous humor from the posterior chamber to the anterior chamber. Pressure builds in the posterior chamber, bowing the iris forward (iris bombé) and occluding the trabecular meshwork. (D) A neovascular membrane has grown over the surface of the iris, smoothing the iris folds and crypts. Myofibroblasts within the neovascular membrane cause the membrane to contract and become apposed to the trabecular meshwork (peripheral anterior synechiae). The outflow of aqueous humor is blocked, and the intraocular pressure becomes elevated.

in the open angle. The underlying cause of the increased resistance is unknown. Genetic factors are suspected to play a role and GWAS has identified 40 genes associated with open angle glaucoma.

- In *angle-closure glaucoma*, the peripheral zone of the iris adheres to the trabecular meshwork and physically impedes the egress of aqueous humor from the eye.

Both open-angle and angle-closure glaucoma can be subclassified into primary and secondary types.

- In *primary open-angle glaucoma*, the most common form, the angle is open, and few changes are apparent structurally.
- There are multiple causes of *secondary open-angle glaucoma*. *Pseudoexfoliation glaucoma*, the most common form of secondary open-angle glaucoma, is associated with the deposition of fibrillar material of varying composition throughout the anterior segment. In addition to deposition in the anterior chamber, fibrillar material is deposited around blood vessels in connective tissue and many visceral organs such as the liver, kidney, and gallbladder.
- Particulate material such as high-molecular-weight lens proteins produced by dissolution of the lens (*phacolytic glaucoma*), senescent red cells after trauma (*ghost cell glaucoma*), iris epithelial pigment granules (*pigmentary glaucoma*), and necrotic tumors (*melanolytic glaucoma*) can clog the trabecular meshwork in the presence of an open angle. Elevations in the pressure on the surface of the eye (episcleral venous pressure) in the presence of an open angle also contribute to *secondary open-angle glaucoma*. This type of glaucoma is associated with surface ocular vascular malformations seen in *Sturge-Weber syndrome* or is a consequence of arterIALIZATION of the episcleral veins following a spontaneous or traumatic carotid-cavernous fistula.
- *Primary angle-closure glaucoma* typically develops in eyes with shallow anterior chambers, often in individuals with hyperopia. Transient apposition of the iris's pupillary margin to the lens's anterior surface may obstruct the flow of aqueous humor through the pupillary aperture (*pupillary block*). Continued production of aqueous humor by the ciliary body elevates pressure in the posterior chamber and may bow the iris periphery forward (*iris bombé*), pushing it against the trabecular meshwork. These anatomic changes provoke a marked elevation in intraocular pressure (Fig. 29.12). Since the crystalline lens is avascular and the lens epithelium receives its nutrition from the aqueous humor, unremitting elevation in intraocular pressure in primary angle-closure glaucoma can damage the lens epithelium. This leads to minute anterior subcapsular opacities that are visible by slit-lamp examination. Although the affected individual might have a normal complement of healthy corneal endothelial cells, sustained elevated intraocular pressure can produce corneal edema and bulbous keratopathy.
- There are many causes of *secondary angle-closure glaucoma*. Contraction of various types of pathologic membranes that form over the surface of the iris can draw the iris over the trabecular meshwork, occluding aqueous outflow. For example, chronic retinal ischemia is associated with upregulation of vascular endothelial growth factor (VEGF) and other proangiogenic factors. The presence of VEGF in the aqueous humor is thought to induce the development of thin, clinically transparent fibrovascular membranes over the surface of the iris. Contraction of myofibroblastic elements in these membranes leads to occlusion of the trabecular meshwork by the iris, producing *neovascular glaucoma* (Fig. 29.12). Necrotic tumors, especially retinoblastomas, can also induce iris neovascularization and

glaucoma. Other mechanisms may also cause secondary angle-closure glaucoma; for example, tumors in the ciliary body can mechanically press the iris onto the trabecular meshwork, closing off the major pathway of aqueous outflow.

KEY CONCEPTS

Anterior Segment

- The term cataract describes opacities of the crystalline lens that may be congenital or acquired.
- Glaucoma describes conditions characterized by distinctive changes in the visual field and the size and shape of the optic nerve cup and usually by an elevation in intraocular pressure.
- Glaucoma may develop in the context of either an open or closed angle. Open-angle and angle-closure glaucomas are further subclassified into primary and secondary types.

UVEA

The iris, choroid, and ciliary body constitute the uvea. The choroid is among the most richly vascularized tissues in the body.

Uveitis

Uveitis can be applied to any inflammation in one or more uveal tissues. Thus, the iritis that develops after blunt trauma to the eye or that accompanies a corneal ulcer is technically a form of uveitis. However, in clinical practice, the term *uveitis* is restricted to a diverse group of chronic diseases that may be either a component of a systemic process or localized to the eye. Uveal inflammation may be manifest principally in the anterior segment (e.g., in *juvenile rheumatoid arthritis*) or may affect both the anterior and posterior elements. The complications of chronic anterior segment inflammation were discussed earlier; here we focus on the effects of uveal inflammation on the posterior segment of the eye. As will be described briefly, uveitis is frequently accompanied by retinal pathology. Uveitis may be caused by infectious agents (e.g., *Pneumocystis carinii*), may be idiopathic (e.g., sarcoidosis), or may be autoimmune in origin (sympathetic ophthalmia).

- *Granulomatous uveitis* is a common complication of sarcoidosis (Chapter 15). In the anterior segment it gives rise to an exudate that evolves into “mutton-fat” keratic precipitates described earlier. In the posterior segment, sarcoid may involve the choroid and retina. Thus, granulomas may be seen in the choroid. Retinal pathology is characterized by perivascular inflammation; this is responsible for the well-known ophthalmoscopic sign of “candle wax drippings.” Conjunctival biopsy can detect granulomatous inflammation and confirm the diagnosis of ocular sarcoid.
- Numerous *infectious processes* can affect the choroid or the retina. Inflammation in one compartment is typically associated with inflammation in the other. Retinal *toxoplasmosis* is usually accompanied by uveitis and sometimes scleritis. Individuals with AIDS may develop cytomegalovirus retinitis and uveal infection with organisms such as *Pneumocystis*, or mycobacterial choroiditis.
- *Sympathetic ophthalmia* is an example of noninfectious uveitis limited to the eye. This condition is characterized by bilateral granulomatous inflammation typically affecting all components of the uvea (*panuveitis*). Sympathetic ophthalmia, which blinded young Louis Braille, may complicate a penetrating eye injury. In the injured eye, retinal antigens sequestered from the immune system may gain access to the conjunctiva's lymphatics, thus setting up a delayed hypersensitivity reaction that affects the injured

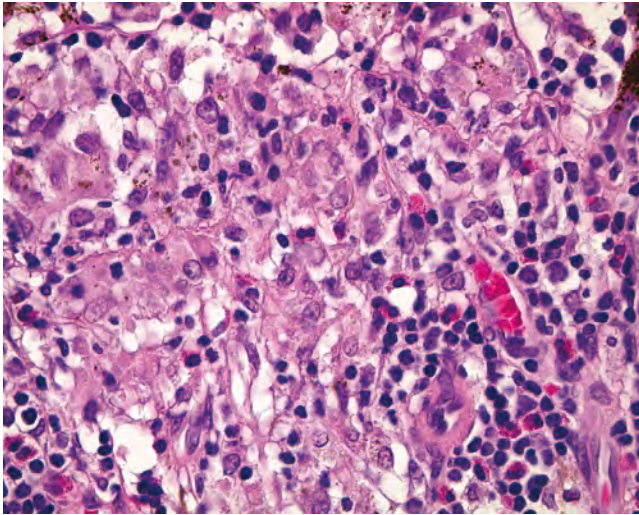


Fig. 29.13 Sympathetic ophthalmia. The granulomatous inflammation depicted here was identified diffusely throughout the uvea. The uveal granulomas may contain melanin pigment and may be accompanied by eosinophils.

eye as well as the contralateral, noninjured eye. The condition may develop from 2 weeks to many years after injury. Enucleation of a blind eye (which may be the sympathizing eye rather than the previously injured eye) may yield diagnostic findings. Histologic examination shows diffuse granulomatous inflammation of the uvea (choroid, ciliary body, and iris). Plasma cells are typically absent, but eosinophils may be identified in the infiltrate (Fig. 29.13). Sympathetic ophthalmia is treated with systemic immunosuppressive agents.

Neoplasms

The most common intraocular malignancy of adults is metastasis to the uvea, typically to the choroid (primarily from lung cancer in males and breast cancer in females). The occurrence of metastases to the eye is associated with extremely short survival, and treatment of ocular metastases, usually by radiotherapy, is palliative.

Uveal Nevi and Melanomas

Uveal melanoma is the most common primary intraocular malignancy in adults. In the United States, these tumors account for approximately 5% of melanomas and have an age-adjusted incidence of 5.1 per 1 million per year. Uveal nevi, especially choroidal nevi, are relatively common, affecting an estimated 5% of the population of European descent.

Epidemiology and Pathogenesis

Unlike cutaneous melanoma, the incidence of uveal melanoma has remained stable over many years, and there is no clear link between exposure to ultraviolet light and risk. In line with this observation, sequencing of tumor genomes has revealed that the molecular pathogenesis of uveal melanoma is distinct from that of cutaneous melanoma. Unlike cutaneous and conjunctival melanomas, *BRAF* mutations do not appear to play a role in the pathogenesis of uveal melanomas. The most critical oncogenes in uveal melanoma are *GNAQ* and *GNA11*, which encode G protein-coupled receptors. Roughly 85% of uveal melanomas harbor a gain-of-function mutation in one of these genes, which activate pathways that promote proliferation, such as the mitogen-activated protein kinase (MAPK) pathway (Chapter 7). Notably, *uveal nevi* are also

associated with *GNAQ* and *GNA11* mutations yet rarely transform into melanoma, indicating that additional genetic events are required to develop uveal melanoma. One such event is the loss of chromosome 3, which appears to be selected for because it leads to the deletion of *BAP1*, a tumor suppressor gene on chromosome 3 that encodes a deubiquitinating enzyme. *BAP1* is a component of protein complexes that place repressive marks on chromatin that lead to gene silencing; thus, uveal melanoma has joined the increasing list of cancers in which epigenetic alterations appear to have a central role in tumor pathogenesis (Chapter 7). Germline mutations in *BAP1* predispose patients to uveal melanomas, renal cell carcinomas, and malignant mesothelioma.

MORPHOLOGY

Histologically, uveal melanomas may contain two types of cells, spindle and epithelioid, in various proportions (Fig. 29.14). **Spindle cells** are fusiform in shape, whereas **epithelioid cells** are spherical and have greater cytologic atypicity. Like cutaneous melanomas, many tumor-infiltrating lymphocytes may be seen in some cases. An unusual feature commonly seen is the presence of looping slit-like spaces lined by laminin that surround packets of tumor cells. These spaces (which are not blood vessels) connect to blood vessels and serve as extravascular conduits for the transport of plasma and possibly blood. In vitro studies and examination of human tissues suggest that these unusual growth patterns are promoted by tumor cells through a process termed **vasculogenic mimicry**.

Uveal melanomas spread almost exclusively by a hematogenous route (the exception being the unusual case of melanoma that extends through the sclera and invades the conjunctiva, thereby gaining access to conjunctival lymphatics). Most uveal melanomas disseminate first to the liver, a classic example of a tumor-specific tropism for a metastatic site.

Clinical Features

Most uveal melanomas are incidental findings or present with visual symptoms, which may be related to retinal detachment or glaucoma. The prognosis of choroidal and ciliary body melanomas is related to (1) size (in contrast to cutaneous melanoma, the lateral extent of the tumor rather than tumor depth is related to an adverse outcome); (2) cell type (tumors containing epithelioid cells have a worse prognosis than those containing exclusively spindle cells); and (3) proliferative index. Gene expression profiling may help stratify patients into categories for developing metastatic disease.

There is no difference in survival between tumors treated by removal of the eye (enucleation) and those receiving eye-sparing radiotherapy, which is the treatment of choice. Melanomas situated exclusively in the iris tend to follow a relatively indolent course, whereas melanomas of the ciliary body and choroid are more aggressive.

Although the 5-year survival rate is approximately 80%, the cumulative melanoma mortality rate is 40% at ten years, increasing 1% annually. Metastases may appear “out of the blue” many years after treatment, making uveal melanoma a prime candidate for investigating the phenomenon of tumor dormancy. Targeted therapies such as MAPK inhibitors have shown some encouraging responses in clinical trials, but currently, there is no effective treatment for metastatic uveal melanoma.

KEY CONCEPTS

Uvea

- Uveitis is restricted to a diverse group of chronic diseases that may be either a component of a systemic disease or localized to the eye.
- Sarcoidosis is an example of a systemic condition that may produce granulomatous uveitis, and sympathetic ophthalmia may have bilateral

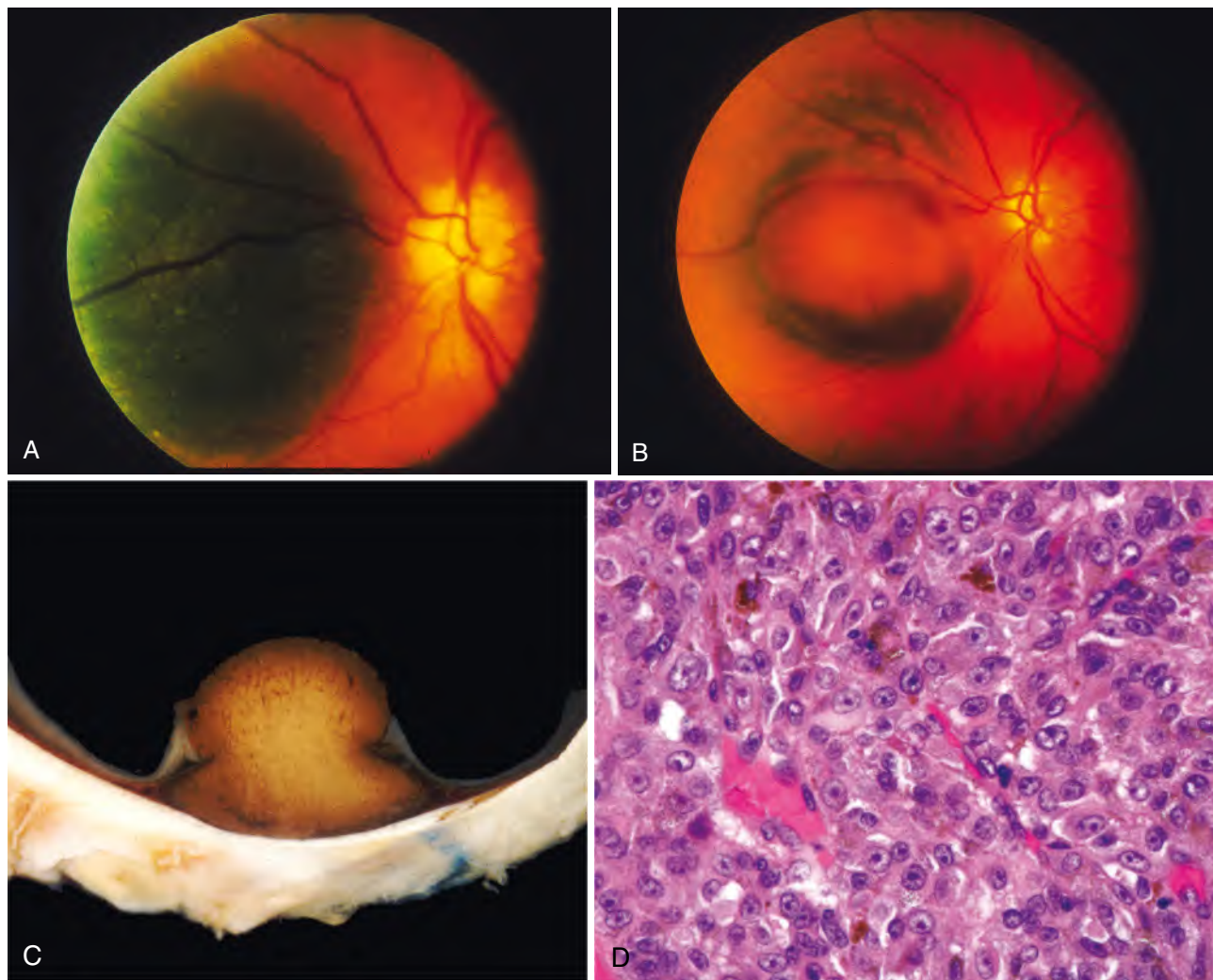


Fig. 29.14 Uveal melanoma. (A) Fundus photograph from an individual with a relatively flat pigmented lesion of the choroid near the optic disc. (B) Fundus photograph of the same individual several years later; the tumor has grown and has ruptured through the Bruch membrane. (C) Gross photograph of a choroidal melanoma that has ruptured the Bruch membrane. The overlying retina is detached. (D) Epithelioid melanoma cells are associated with an adverse outcome. (A–C, From Folberg R. *Pathology of the Eye—An Interactive CD-ROM Program*. Mosby; 1996.)

granulomatous inflammation as a possible consequence of penetrating injury to one eye.

- The most common intraocular tumor in adults is metastasis to the eye.
- The most common primary intraocular tumor in adults is uveal melanoma.
- Uveal melanoma disseminates hematogenously, and the first evidence of metastasis is typically detected in the liver.
- Uveal melanoma shows marked differences in epidemiologic risk factors and driver mutations compared with cutaneous melanoma.

RETINA AND VITREOUS

The neurosensory retina, like the optic nerve, is derived embryologically from the diencephalon. The retina, therefore, responds to injury through gliosis. As in the brain, there are no lymphatics. The architecture of the retina accounts for the ophthalmoscopic appearance of various ocular disorders. Hemorrhages in the nerve fiber layer of the

retina are oriented horizontally and appear as streaks or “flames”; the external retinal layers are oriented perpendicular to the retinal surface, and hemorrhages in these outer layers appear as dots (the tips of cylinders). Exudates tend to accumulate in the outer plexiform layer of the retina, especially in the macula (Fig. 29.15).

The *retinal pigment epithelium* (RPE), like the retina, is derived embryologically from the primary optic vesicle, an outpouching of the brain. Separation of the neurosensory retina from the RPE defines a *retinal detachment*. The RPE has a vital role in maintaining the outer segments of the photoreceptors. Disturbances in the RPE–photoreceptor interface are implicated in hereditary retinal degenerations such as *retinitis pigmentosa*.

The adult vitreous humor is avascular. Incomplete regression of fetal vasculature running through the vitreous humor can produce significant pathology as a retrolental mass (*persistent fetal vasculature*). The vitreous humor can be opacified by hemorrhage from trauma or retinal neovascularization. With age, the vitreous humor

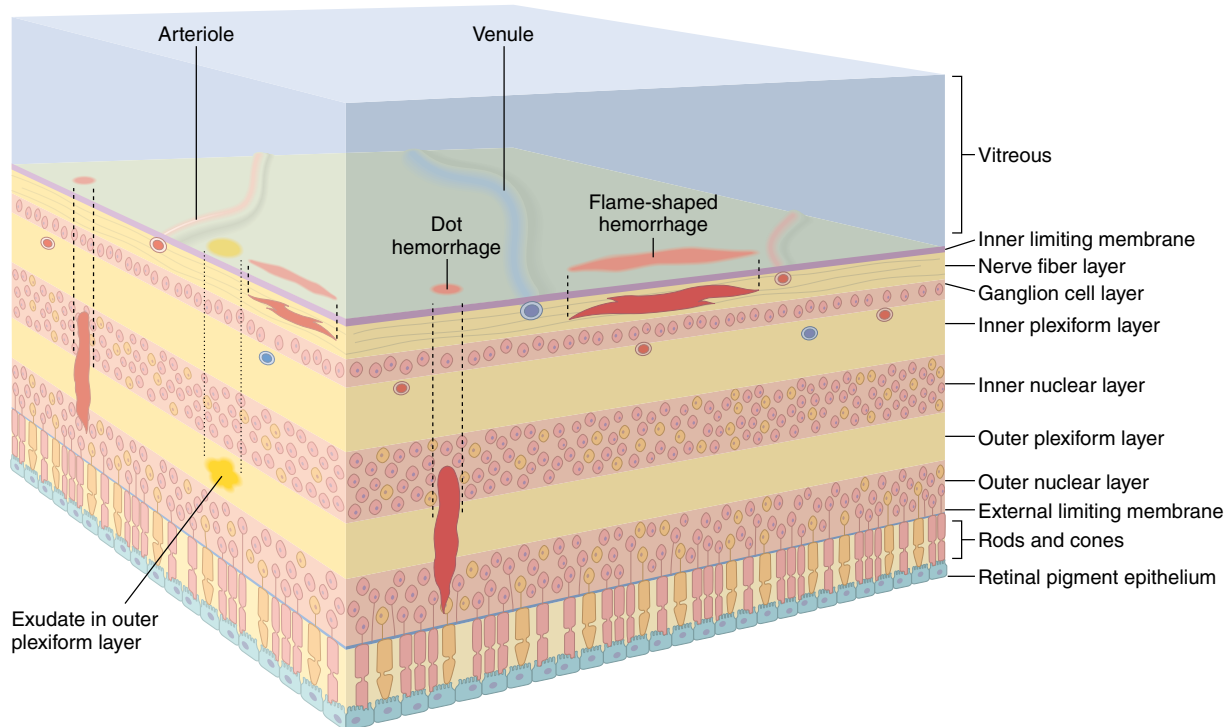


Fig. 29.15 Clinicopathologic correlations of retinal hemorrhages and exudates. The location of the hemorrhage within the retina determines its appearance by ophthalmoscopy. The retinal nerve fiber layer is oriented parallel to the internal limiting membrane, and hemorrhages of this layer appear to be flame-shaped ophthalmoscopically. The deeper retinal layers are oriented perpendicular to the internal limiting membrane, and hemorrhages in this location appear as cross-sections of a cylinder, or “dot” hemorrhages. Exudates that originate from leaky retinal vessels accumulate in the outer plexiform layer.

may liquefy and collapse, creating the visual sensation of “floaters.” Also, with aging, the posterior face of the vitreous humor (i.e., the posterior hyaloid) may separate from the neurosensory retina (*posterior vitreous detachment*). The relationship between the posterior hyaloid and the neurosensory retina plays a crucial role in the evolution of retinal neovascularization and some forms of retinal detachment.

Acute Intraocular Inflammation: Retinitis, Endophthalmitis, and Panophthalmitis

A variety of pathogens can contribute to the development of infectious retinitis. For example, *Candida* may disseminate to the retina hematogenously, especially in users of intravenous drugs or systemic candidemia from other causes. Hematogenous dissemination of pathogens to the retina typically results in multiple retinal abscesses. As was mentioned previously, cytomegalovirus retinitis is a significant cause of visual morbidity in immunocompromised individuals, especially those with AIDS.

Endophthalmitis is classified as *exogenous* (originating in the environment and gaining access to the eye’s interior through a wound as a complication of surgery or penetrating injury) or *endogenous* (delivered to the eye hematogenously, e.g., secondary to pulmonary infections). The retina lines the vitreous cavity, and the retina poorly tolerates suppurative inflammation in the vitreous humor; a few hours may be sufficient to cause irreversible retinal injury. Panophthalmitis is applied to eye inflammation involving the retina, choroid, and sclera extending into the orbit (Fig. 29.16).

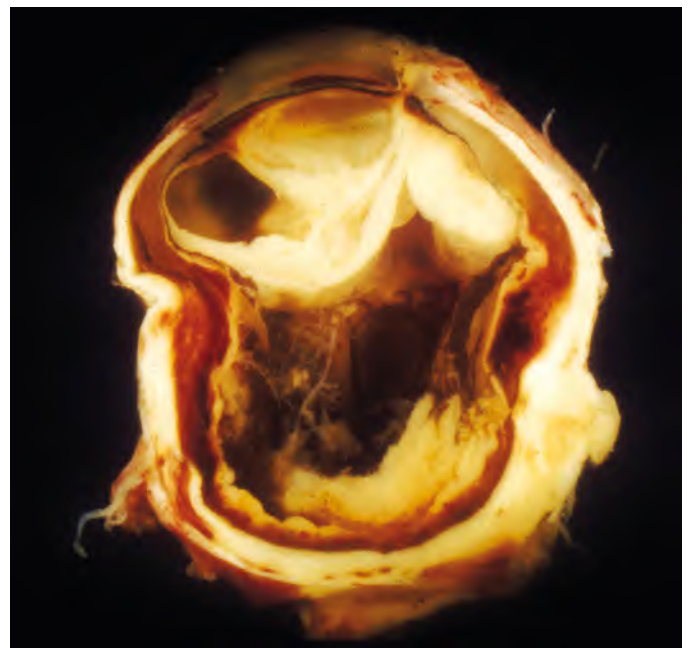
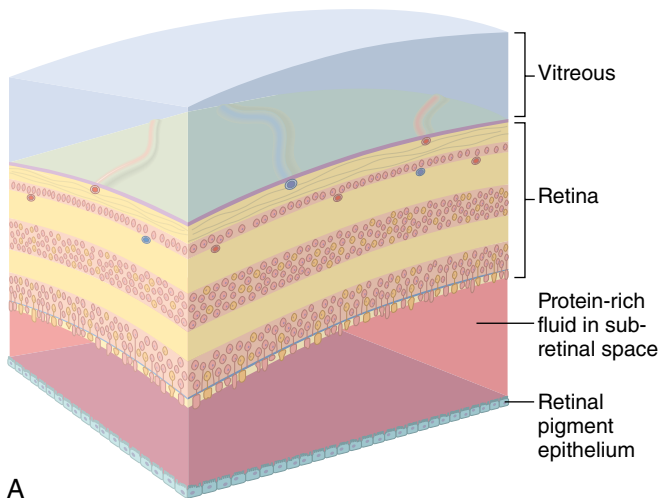
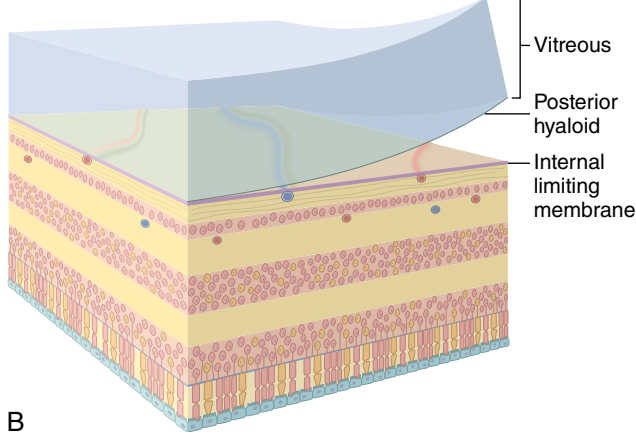


Fig. 29.16 Exogenous panophthalmitis. This eye was removed after a foreign body injury. Suppurative inflammation of the entire orbit has distorted the architecture of the eye. (From Folberg R. The eye. In: Spencer WH, ed. *Ophthalmic Pathology—An Atlas and Textbook*, 4th ed. WB Saunders; 1985.)

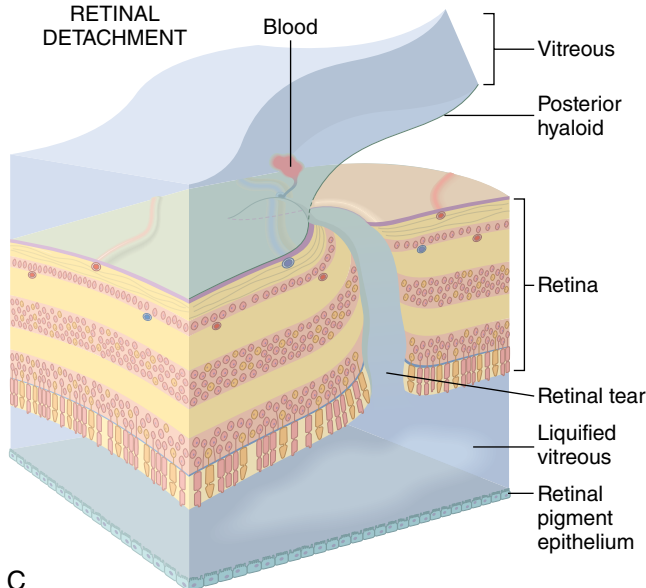
NON-RHEGMATOGENOUS
RETINAL DETACHMENT

A

VITREOUS DETACHMENT



B

RHEGMATOGENOUS
RETINAL DETACHMENT

C

Fig. 29.17 Retinal detachment is defined as the separation of the neurosensory retina from the retinal pigment epithelium (RPE). Retinal detachments are classified broadly as nonrhegmatogenous (without a retinal break) and rhegmatogenous (with a retinal break). (A) In nonrhegmatogenous retinal detachment, the subretinal space is filled with protein-rich exudate. Note that the outer segments of the photoreceptors are missing (see Fig. 29.15 for the orientation of layers). This indicates a chronic retinal detachment, a finding that can be seen in both nonrhegmatogenous and rhegmatogenous detachments. (B) Posterior vitreous detachment involves the separation of the posterior hyaloid from the internal limiting membrane of the retina and is a regular occurrence in the aging eye. (C) If during a posterior vitreous detachment the posterior hyaloid does not separate cleanly from the internal limiting membrane of the retina, the vitreous humor will exert traction on the retina, which will be torn at this point. Liquefied vitreous humor seeps through the retinal defect, separating the retina from the RPE. The photoreceptor outer segments are intact, illustrating an acute detachment.

KEY CONCEPTS

Endophthalmitis

- Endophthalmitis is a term used to describe inflammation of the interior of the eye involving the vitreous humor, and panophthalmitis is the term used to describe inflammation of the interior of the eye that extends into the uvea and sclera.
- Endophthalmitis may originate from infection within the body (endogenous endophthalmitis complicating generalized sepsis) or as a complication of corneal infection or a wound, an accidental injury, or a surgical procedure (exogenous endophthalmitis).

Retinal Detachment

Retinal detachment (separation of the neurosensory retina from the RPE) is broadly classified into rhegmatogenous and nonrhegmatogenous based on the presence or absence of a break in the retina.

- *Rhegmatogenous retinal detachment* is associated with a full-thickness retinal defect. Retinal tears may develop after the vitreous collapses structurally, and the posterior hyaloid exerts traction on points of abnormally strong adhesion to the retinal internal limiting membrane. Liquefied vitreous humor then seeps through the tear and accesses the potential space between the neurosensory retina and the RPE (Fig. 29.17).

Reattachment of the retina to the RPE generally requires relief of vitreous traction through indenting of the sclera by surgical procedures. This can be accomplished by the application of strips of silicon to the surface of the eye (scleral buckling) and possibly by removing vitreous material (vitrectomy). Rhegmatogenous retinal detachment may be complicated by *proliferative vitreoretinopathy* with the formation of epiretinal or subretinal membranes by retinal glial cells or RPE cells.

- *Nonrhegmatogenous retinal detachment* (retinal detachment without a retinal break) may complicate retinal vascular disorders associated with significant exudation and any condition that damages the RPE and permits fluid to leak from the choroidal circulation under the retina. Retinal detachments associated with choroidal tumors and severe hypertension are examples of nonrhegmatogenous retinal detachment.

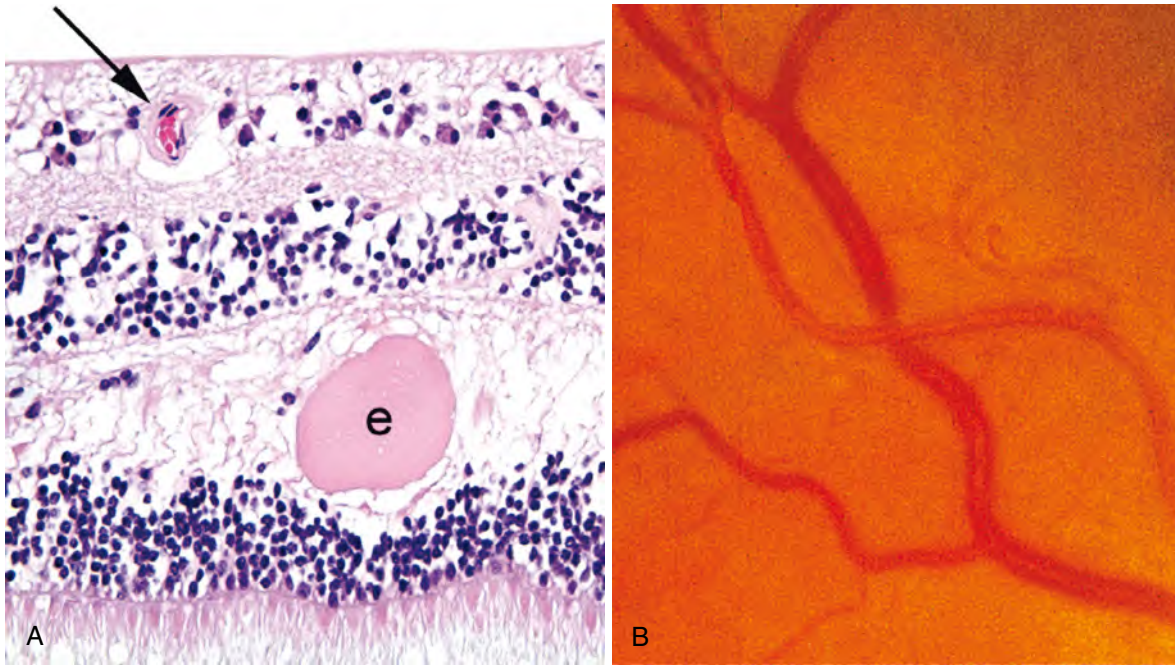


Fig. 29.18 The retina in hypertension. (A) The wall of the retinal arteriole (*arrow*) is thick. Note the exudate (*e*) in the retinal outer plexiform layer. (B) The fundus in hypertension. The diameter of the arterioles is reduced, and the color of the blood column appears to be less saturated (copper wire–like). If the wall of the vessel were thicker still, the degree of red color would diminish such that the vessels might appear clinically to have a “silver wire” appearance. Note that the vein is compressed where the sclerotic arteriole crosses over it. (B, Courtesy Dr. Thomas A. Weingeist, Department of Ophthalmology and Visual Science, University of Iowa, Iowa City.)

Retinal Vascular Disease

Hypertension

Typically, the thin walls of retinal arterioles permit direct visualization of the circulating blood by ophthalmoscopy. In retinal arteriosclerosis, which is associated with long-standing hypertension, the thickened arteriolar wall changes the ophthalmic perception of circulating blood: vessels may appear narrow, and the color of the blood column may vary from bright red to copper and silver depending on the degree of vascular wall thickness (Fig. 29.18A). Retinal arterioles and veins share a common adventitial sheath. Therefore, in pronounced retinal arteriosclerosis, the arteriole may compress the vein at points where both vessels cross (Fig. 29.18B). Venous stasis distal to arteriolar-venous crossing may precipitate occlusions of the retinal vein branches.

In severe hypertension, vessels in the retina and choroid may be injured. Damage to choroidal vessels may produce focal choroidal infarcts, seen clinically as *Elschnig spots*. Damage to the choriocapillaris, the internal layer of the choroidal vasculature, may, in turn, damage the overlying RPE and permit the exudate to accumulate in the potential space between the neurosensory retina and the RPE, thereby producing a retinal detachment. Exudate from damaged retinal arterioles typically accumulates in the outer plexiform layer of the retina (see Fig. 29.18A). The ophthalmoscopic finding of a macular star—a spoke-like arrangement of exudate in the macula in severe hypertension—results from exudate accumulating in the outer plexiform layer of the macula that is oriented obliquely instead of perpendicular to the retinal surface.

Occlusion of retinal arterioles may produce infarcts of the nerve fiber layer of the retina (axons of the retinal ganglion cell layer populate the nerve fiber layer). Axoplasmic transport in the nerve fiber layer is interrupted at the point of axonal damage, and accumulation of mitochondria at the swollen ends of damaged axons creates the histologic illusion of cells (*cytoid bodies*). Collections of cytoid bodies populate the nerve fiber layer infarct, seen ophthalmoscopically as “cotton-wool spots” (Fig. 29.19). Although nerve fiber layer infarcts are described here in the context of hypertension, they may be detected in other forms of retinal occlusive vasculopathy.

Diabetes Mellitus

The eye is profoundly affected by diabetes. The effects of hyperglycemia on the lens and iris have already been mentioned. Thickening of the basement membrane of the epithelium of the pars plicata (anterior part of the ciliary body that produces aqueous humor) is a reliable histologic marker of diabetes in the eye (Fig. 29.20) and is reminiscent of similar changes in the glomerular mesangium. The biochemical changes responsible for vascular changes in diabetes are discussed in Chapter 24. This discussion focuses on the retinal microangiopathy associated with diabetes.

Pathogenesis

Approximately 60% to 80% of patients develop some form of diabetic retinopathy 15 to 20 years after diagnosis. The fundamental lesion of

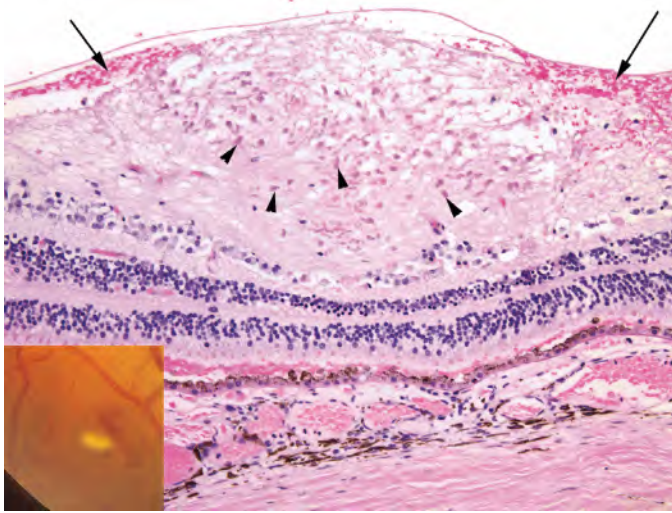


Fig. 29.19 Nerve fiber layer infarct. A “cotton-wool spot” is illustrated in the inset, adjacent to a flame-shaped (nerve fiber layer) hemorrhage. The histology of a cotton-wool spot, an infarct of the nerve fiber layer of the retina, is illustrated in the photomicrograph. Numerous red to pink cytotoid bodies (*arrowheads*) occupy a focal swelling of the nerve fiber layer. Hemorrhage (*arrows*) surrounding the nerve fiber layer infarct is a variable and inconsistent finding, as illustrated here. (Fundus photograph, Courtesy Dr. Thomas A. Weingeist, Department of Ophthalmology and Visual Science, University of Iowa, Iowa City)

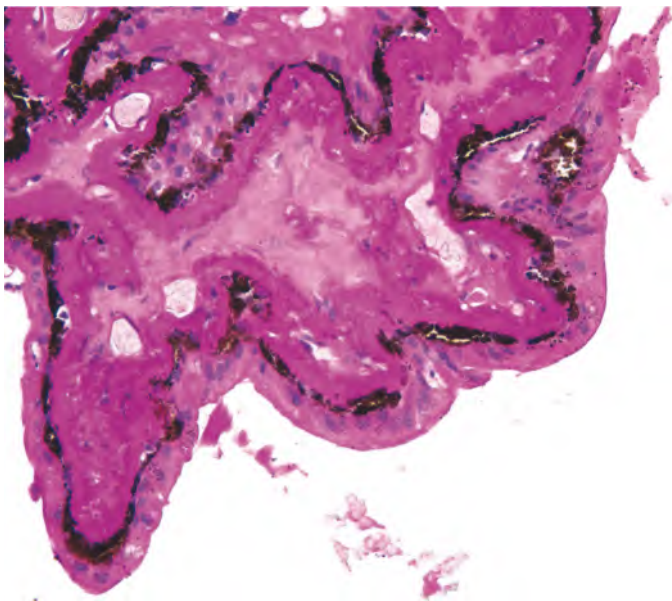


Fig. 29.20 The ciliary body in chronic diabetes, periodic acid-Schiff stain. Note the massive thickening of the basement membrane of the ciliary body epithelia, reminiscent of changes in the mesangium of the renal glomerulus.

retinopathy neovascularization is attributable to hypoxia-induced expression of VEGF in retina. The retinal vasculopathy of diabetes can be classified into nonproliferative and proliferative diabetic retinopathy. These are illustrated in [Fig. 29.21](#).

MORPHOLOGY

Nonproliferative diabetic retinopathy includes a spectrum of changes resulting from structural and functional abnormalities of retinal vessels (i.e., confined beneath the internal limiting membrane of the retina). As with diabetic microangiopathy elsewhere, the basement membrane of retinal blood vessels is generally **thickened**. In addition, the number of pericytes relative to endothelial cells diminishes. **Microaneurysms** are a prominent manifestation of diabetic microangiopathy. They are typically smaller than the resolution of direct ophthalmoscopes, and findings customarily described as microaneurysms by ophthalmoscopy may be retinal microhemorrhages. Structural changes in the retinal microcirculation have been associated with a physiologic breakdown in the blood-retinal barrier. Recall that VEGF was initially called vascular permeability factor. Thus, the retinal microcirculation in diabetics may be exceptionally leaky, giving rise to **macular edema**, a common cause of visual loss in these patients. The vascular changes may also produce **exudates** accumulating in the outer plexiform layer. Although the retinal microcirculation is often hyperpermeable, it is also subject to micro-occlusion. Both vascular incompetence and micro-occlusions can be visualized clinically after intravenous fluorescein injection. Nonperfusion of the retina due to the microcirculatory change described earlier is associated with the upregulation of VEGF and intraretinal angiogenesis (located beneath the internal limiting membrane of the retina).

The neovascularization of proliferative diabetic retinopathy is defined by the appearance of new vessels sprouting on the surface of either the optic nerve head (neovascularization of the disc or NVD) or the retina's surface (neovascularization elsewhere or NVE) ([Fig. 29.22C](#)). The term “retinal neovascularization” is applied only when the newly formed vessels breach the internal limiting membrane of the retina. The quantity and location of retinal neovascularization guide the ophthalmologist in treating proliferative diabetic retinopathy. The web of newly formed vessels is called a neovascular membrane: angiogenic vessels with or without a substantial supportive fibrous or glial stroma ([Fig. 29.22B](#)).

If the vitreous humor has not detached and the posterior hyaloid is intact, neovascular membranes extend along the potential plane between the retinal internal limiting membrane and the posterior hyaloid. If vitreous humor later separates from the internal limiting membrane of the retina (**posterior vitreous detachment**), there may be massive hemorrhage ([Fig. 29.21](#)) from the disrupted neovascular membrane. In addition, scarring associated with the organization of the retinal neovascular membrane may wrinkle the retina, disrupting the orientation of retinal photoreceptors and producing visual distortion. It may exert traction on the retina, separating it from the RPE (retinal detachment). **Traction retinal detachment** may begin as a nonrhegmatogenous detachment, but severe traction may tear the retina, producing a rhegmatogenous traction detachment.

Retinal neovascularization may be accompanied by the development of a neovascular membrane on the iris surface, presumably secondary to increased levels of VEGF in the aqueous humor.

Contraction of the neovascular iris membrane may lead to adhesions between the iris and trabecular meshwork (anterior synechiae), thus occluding a significant pathway for aqueous outflow and thereby contributing to the elevation of the intraocular pressure (**neovascular glaucoma**).

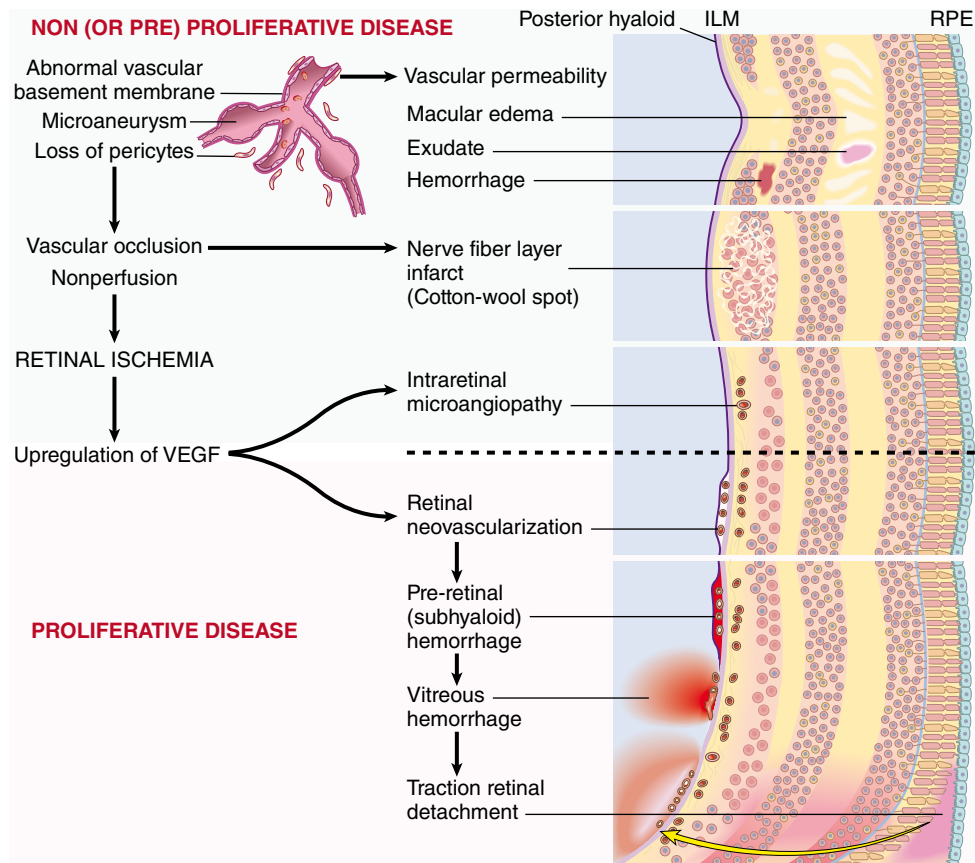


Fig. 29.21 Progression of diabetic retinopathy. In non- or pre-proliferative disease, the changes are intraretinal. The presence of a neovascular membrane that breaches the retinal internal limiting membrane defines proliferative diabetic retinopathy.

Ablating the nonperfused retina by laser photocoagulation or cryopexy triggers regression of both retinal and iris neovascularization, emphasizing the central role that retinal hypoxia has in these disorders. The injection of VEGF inhibitors into the vitreous has been used to treat diabetic macular edema and retinal neovascularization, a successful example of how knowledge of the molecular pathogenesis of a condition may evolve into a successful therapeutic strategy.

Retinopathy of Prematurity (Retrolental Fibroplasia)

In premature or low-birth-weight infants treated with oxygen, immature retinal vessels in the temporal retinal periphery constrict, rendering retinal tissue distal to this zone ischemic. Retinal ischemia upregulates proangiogenic factors such as VEGF, leading to retinal angiogenesis. Contraction of the resulting peripheral retinal neovascular membrane may “drag” the temporal aspect of the retina toward the peripheral zone, displacing the macula (situated temporally to the optic nerve) laterally. Moreover, neovascular membrane contraction may create sufficient force to cause retinal detachment. The use of VEGF inhibition to prevent this condition is an area of active investigation.

Sickle Retinopathy, Retinal Vasculitis, and Radiation Retinopathy

Retinopathy affecting individuals with sickle hemoglobinopathies (Chapter 14) has been divided into two types that roughly parallel those seen in diabetic retinopathy: nonproliferative (intraretinal angiopathic changes) and proliferative (retinal neovascularization). The final common pathway is vascular occlusion. Low oxygen tension within the blood vessels in the retinal periphery results in red cell sickling and microvascular occlusions. In the nonproliferative form (which occurs in individuals with hemoglobin SS and SC genotypes), vascular occlusions are thought to contribute to preretinal, intraretinal, and subretinal hemorrhages. The resolution of these hemorrhages may give rise to several ophthalmoscopically visible changes known as *salmon patches*, *iridescent spots*, and *black sunburst lesions*. Organization of preretinal hemorrhage may result in retinal traction and retinal detachment. Vascular occlusions may also contribute to angiogenesis secondary to the upregulation of VEGF and basic fibroblast growth factor. This can give rise to areas of florid neovascularization in the retinal periphery, described clinically as “sea fans.”

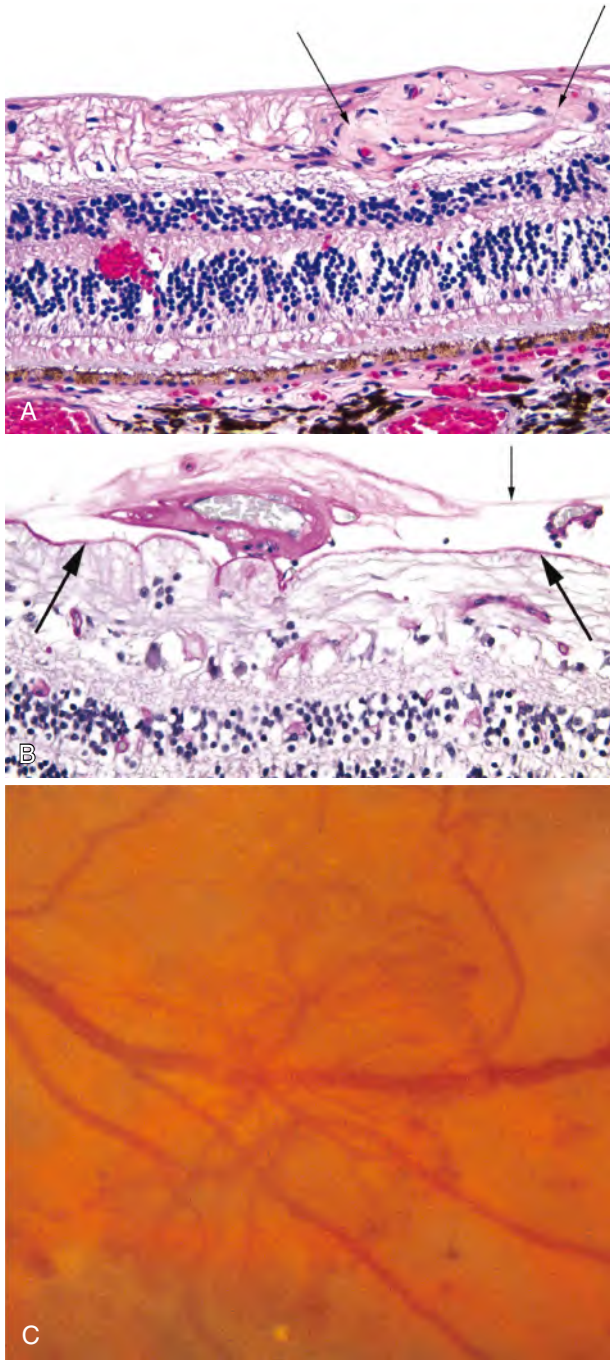


Fig. 29.22 The retina in diabetes. (A) A tangle of abnormal vessels lies just beneath the internal limiting membrane of the retina on the right half of the photomicrograph (*between arrows*). This is an example of intraretinal angiogenesis known as intraretinal microangiopathy (IRMA). Note the retinal hemorrhage in the outer plexiform layer in the left half. The ganglion cell layer and the nerve fiber layer, the axons of the ganglion cells, are absent. The rarefied space beneath the internal limiting membrane to the left of the focus of IRMA consists mainly of elements of retinal glial (Müller) cells. The absence of the ganglion cell and nerve fiber layers is a hallmark of glaucoma. The chronic diabetes in this individual was complicated by iris neovascularization and secondary angle-closure glaucoma (neovascular glaucoma). (B) In this section, stained by periodic acid-Schiff, the internal limiting membrane is indicated by the *thick arrows*, and the posterior hyaloid of the vitreous is indicated by the *thin arrow*. In the potential space between these two landmarks, the vessels to the left of the *thin arrow* are invested with a fibrous-glial stroma. They would appear ophthalmoscopically as a white neovascular membrane. The thin-walled vessel to the right of the *thin arrow* is not invested with connective tissue. A posterior vitreous detachment in an eye such as this might exert traction on these new vessels and precipitate a massive vitreous hemorrhage. (C) The ophthalmoscopic view of retinal neovascularization (known clinically as neovascularization “elsewhere” in contrast to neovascularization of the optic.

Neovascularization also occurs in other clinical settings, such as peripheral retinal vasculitis and in irradiation used to treat intraocular tumors. The feature common to these conditions is damage to retinal vessels, producing zones of retinal ischemia that trigger retinal angiogenesis and its complications, hemorrhage and traction, which in turn may cause detachment.

Retinal Artery and Vein Occlusions

Occlusion of the retinal artery or its branches may lead to retinal infarction and blindness. Thromboembolism is the most common cause. For example, the lumen of the central retinal artery can be narrowed by atherosclerosis, thus predisposing to thrombosis. Emboli to the central retinal artery can originate from thrombi in the heart or

from ulcerated atheromatous plaques in the carotid arteries. Total occlusion of a retinal artery branch can produce a segmental retinal infarct. With the sudden cessation of blood supply, the retina swells acutely (similar to the brain, from which the retina is derived) and becomes optically opaque. By ophthalmoscopy, the fundus in the affected area appears white instead of red or orange because the retinal opacity blocks the view of the richly vascular choroid.

Total occlusion of the central retinal artery can produce a diffuse infarct of the retina. While the infarcted retina becomes opaque, the fovea and foveola are thin; therefore, the choroid’s normal orange-red is visible in the fovea and foveola and highlighted by the surrounding opaque retina—the origin of the *cherry-red spot* of central retinal artery occlusion.

Cherry-red spots can also be seen in rare storage diseases such as *Tay-Sachs* and *Niemann-Pick diseases*. The storage material accumulates in retinal ganglion cells: the ganglion cell layer surrounding the fovea is thick, but there are no ganglion cells in the center of the macula, the fovea. Thus, the fovea is relatively transparent to the underlying choroidal vasculature, appearing as a red spot surrounded by relatively opaque retina (Fig. 29.23).

Retinal vein occlusion may occur with or without ischemia. In ischemic retinal vein occlusion, VEGF and other proangiogenic factors are upregulated in the retina, leading to neovascularization of the retina and surface of the optic nerve head as well as neovascularization of the iris and subsequent angle-closure glaucoma. Hemorrhages, exudates, and macular edema may complicate nonischemic retinal vein occlusion.

Age-Related Macular Degeneration

Age-related macular degeneration (AMD) occurs in two forms, dry and wet, distinguished by the presence of neoangiogenesis in the wet form and its absence in the dry form. Advancing age and genetic predisposition are factors. The cumulative incidence of AMD in individuals 75 years of age and older is 8%, and with increasing longevity AMD is becoming a significant health problem.

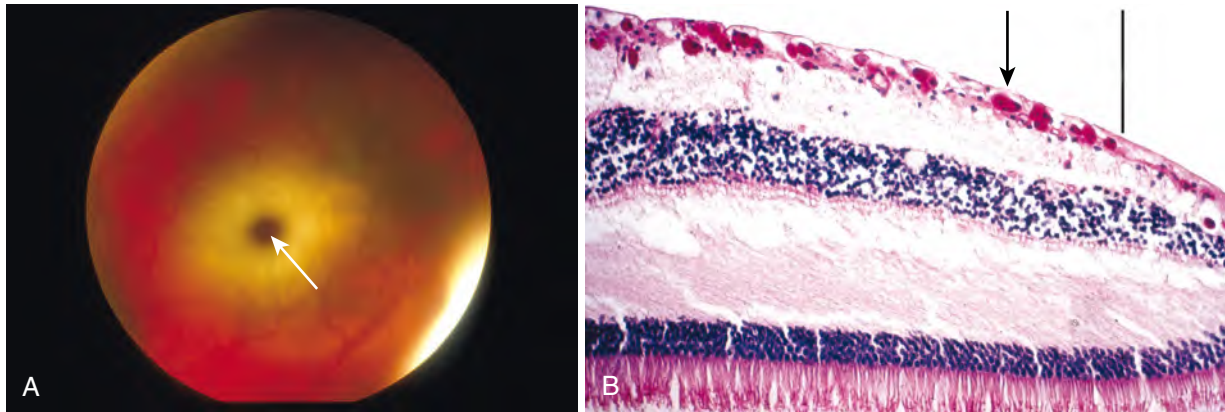


Fig. 29.23 Cherry-red spot in Tay-Sachs disease. (A) Fundus photograph of the cherry-red spot (*arrow*) in Tay-Sachs disease. (B) Photomicrograph of the macula in an individual with Tay-Sachs disease, stained with periodic acid-Schiff to highlight the accumulation of ganglioside material (*arrow*) in the retinal ganglion cells. The presence of ganglion cells filled with gangliosides outside the fovea blocks the transmission of the normal orange-red color of the choroid. The absence of ganglion cells within the fovea (to the right of the *vertical bar*) permits the normal orange-red color to be visualized, accounting for the so-called cherry-red spot. (A, Courtesy Dr. Thomas A. Weingeist, Department of Ophthalmology and Visual Science, University of Iowa, Iowa City. B, From the Armed Forces Institute of Pathology teaching collection.)

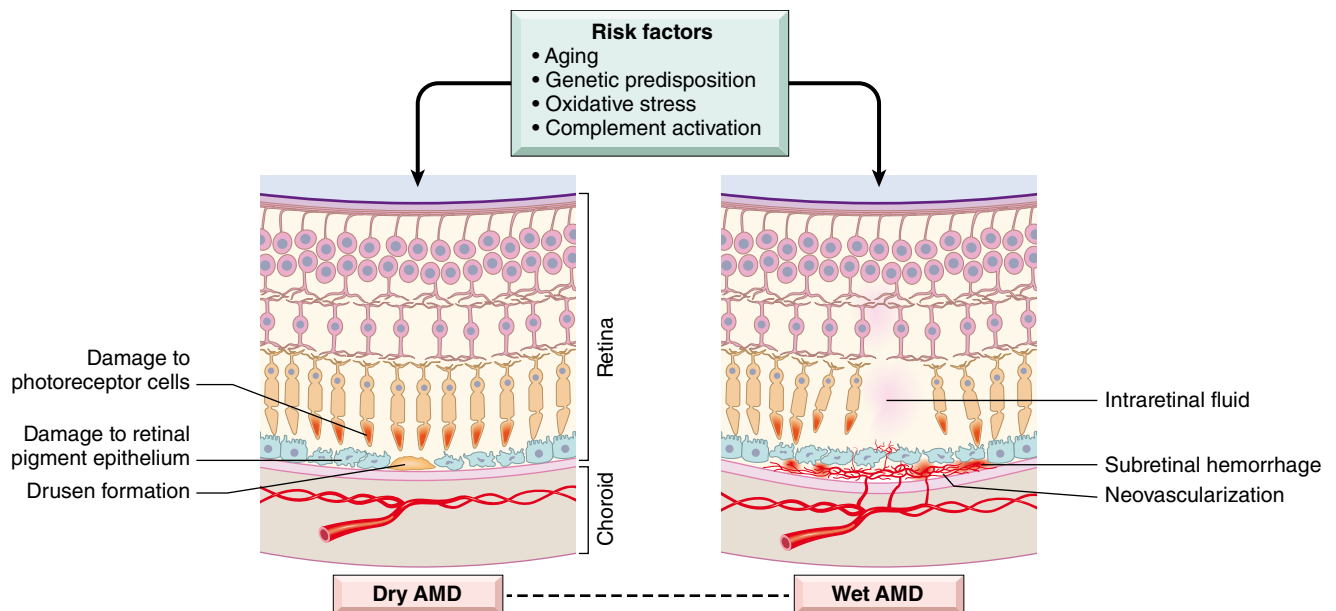


Fig. 29.24 Age-related macular degeneration (AMD). This schematic illustrates the differences between dry and wet AMD. Dry AMD may progress to wet AMD, and in some patients dry and wet AMD may coexist.

Pathogenesis

Understanding the pathogenesis of AMD requires appreciation of the existence of a structural and functional unit comprised of the RPE, Bruch membrane (which contains the basement membrane of the RPE), and the innermost layer of the choroidal vasculature, the choriocapillaris. Disturbance in any component of this “unit” affects the health of the overlying photoreceptors, producing vision loss. With this background, we can discuss the pathogenesis of dry and wet AMD (Fig. 29.24).

Atrophic or “dry” AMD is characterized by diffuse or discrete deposits in the Bruch membrane (*drusen*) and geographic atrophy of the RPE. Geographic atrophy is associated with chronic inflammation due to activation of the complement system that leads to the loss of photoreceptors,

retinal pigment epithelium (RPE), and the underlying choriocapillaris. Loss of vision may be severe in these individuals.

Several genes, especially *CFH* (complement factor H), complement factor I (CFI), and other complement regulatory genes, are believed to be involved in the pathogenesis of this condition. Polymorphisms in genes that regulate the complement system, are associated with an increased risk of developing AMD. These studies suggest that AMD may stem from excess complement activity, which inflicts injury on photoreceptor cells by mechanisms not entirely clear (Fig. 29.25). Environmental exposures such as cigarette smoking may also increase the risk of AMD, especially in genetically predisposed individuals. Therapy targeting complement may slow or halt the progression of

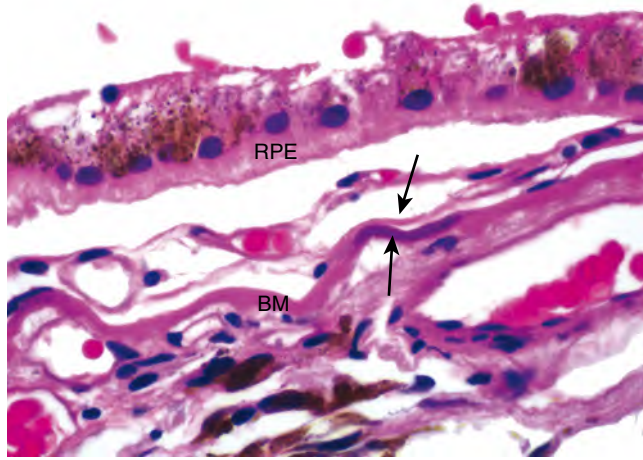


Fig. 29.25 “Wet” age-related macular degeneration. A neovascular membrane is positioned between the retinal pigment epithelium (RPE) and Bruch membrane (BM). Note the blue discoloration of BM to the label’s right, indicating focal calcification (arrows).

geographic atrophy. A regenerative approach to restore vision may include replacing diseased RPE cells with stem cells.

Neovascular or “wet” AMD (also known as *exudative AMD*) is characterized by *choroidal neovascularization*, defined by the presence of vessels that presumably originate from the choriocapillaris and penetrate through the Bruch membrane beneath the RPE (Fig. 29.25). This neovascular membrane may break through the RPE and become situated directly beneath the neurosensory retina. The vessels in this membrane may leak, and the exuded blood may be organized by RPE cells into macular scars. Occasionally, these vessels are the source of hemorrhage, leading to the localized suffusion of blood that may be mistaken clinically for an intraocular neoplasm, or give rise to diffuse vitreous hemorrhage. Currently, the mainstay of treatment for neovascular AMD is the injection of VEGF antagonists into the vitreous of the affected eye.

Dry AMD is not necessarily a precursor to wet AMD. Dry and wet forms may coexist and each form develops through a distinct series of events (Fig. 29.24).

Other Retinal Degenerations

Retinitis Pigmentosa

Retinitis pigmentosa is a degenerative disease associated with loss of photoreceptor cells resulting from mutations that affect rods and cones or RPE. It can cause varying degrees of visual impairment, including, in some cases, total blindness. The term “*retinitis pigmentosa*” is an unfortunate relic of a time when these disorders were incorrectly presumed to be inflammatory. They may be inherited as X-linked recessive, autosomal recessive, or autosomal dominant traits (the age of onset correlates with the inheritance pattern, with autosomal dominant retinitis pigmentosa appearing later in life). Retinitis pigmentosa may be part of a syndrome, such as the *Bardet-Biedl syndrome*, *Usher syndrome*, or *Refsum disease*, or may develop in isolation (nonsyndromic retinitis pigmentosa).

Retinitis pigmentosa is linked to mutations affecting *RHO*, *USH2A*, *RGPR*, and *EYS* genes, all of which regulate the functions of either the photoreceptor cells or the RPE. Symptoms appear as rods and cones die by apoptosis. Cone cell death is partly mediated through the activation of RIP kinase. Loss of rods may lead to early night blindness and constricted visual fields. As cones are lost, central visual acuity may be affected. Retinal atrophy is accompanied by constriction of retinal vessels

and optic nerve head atrophy (“waxy pallor” of the optic disc) and the accumulation of retinal pigment around blood vessels, thus accounting for the “pigmentosa” in the disease name.

Retinal Neoplasms

Retinoblastoma

Retinoblastoma is the most common primary intraocular malignancy in children. The molecular genetics of retinoblastoma are discussed in detail in Chapter 7. Recall that in approximately 40% of cases, retinoblastoma occurs in individuals who inherit a germline mutation of one *RB* allele. Retinoblastoma arises when the retinal progenitor sustains a second somatic mutation, and *RB* gene function is lost. In sporadic cases, both *RB* alleles are lost by somatic mutations. Retinoblastomas arising in those with germline mutations are often bilateral. In addition, they may be associated with pinealoblastoma (“trilateral” retinoblastoma), which has a dismal outcome. In a small proportion of cases, mutations in *RB1* are absent and amplification of the *NMYC* proto-oncogene is observed instead.

MORPHOLOGY

The pathology of retinoblastoma, both hereditary and sporadic types, is identical. Tumors may contain both undifferentiated and differentiated elements (Fig. 29.26). The former appear as collections of small, round cells with hyperchromatic nuclei. In well-differentiated tumors, there are **Flexner-Wintersteiner rosettes** and fleurettes reflecting photoreceptor differentiation. It should be noted, however, that the degree of tumor differentiation does not appear to be associated with the prognosis. Viable tumor cells encircle tumor blood vessels with zones of necrosis typically found in relatively avascular areas, illustrating the dependence of retinoblastoma on its blood supply. Focal zones of dystrophic calcification are characteristic of retinoblastoma.

To preserve vision and eradicate the tumor, many ophthalmic oncologists attempt to reduce tumor burden by administration of chemotherapy, including selective delivery of the drug to the eye through the ophthalmic artery; after chemoreduction, tumors may be obliterated by laser treatment or cryopexy. Retinoblastoma tends to spread to the brain and bone marrow and seldom disseminates to the lungs. Prognosis is adversely affected by extraocular extension, invasion along the optic nerve, and massive choroidal invasion. A closely related proliferation, retinocytoma or retinoma, appears to be a precursor of full-blown retinoblastoma. In line with this idea, the appearance of retinoblastoma in one eye and retinocytoma in the other eye is sometimes seen in heritable retinoblastoma.

Retinal Lymphoma

Primary retinal lymphoma is an aggressive tumor that characteristically involves the two retinal layers derived from the brain, the neurosensory retina and the RPE. Primary intraocular lymphoma tends to occur in older individuals and may mimic uveitis clinically. Most are diffuse large B-cell lymphomas (Chapter 13). Spread to the brain commonly occurs via the optic nerve. The diagnosis requires demonstration of lymphoma cells in vitreous aspirates.

KEY CONCEPTS

Retina

- Retinal detachment, a separation of the neurosensory retina from the RPE, may be the consequence of a break in the retina (rhegmatogenous retinal detachment) or may develop without a retinal break because of pathology within or beneath the retina (nonrhegmatogenous retinal detachment).

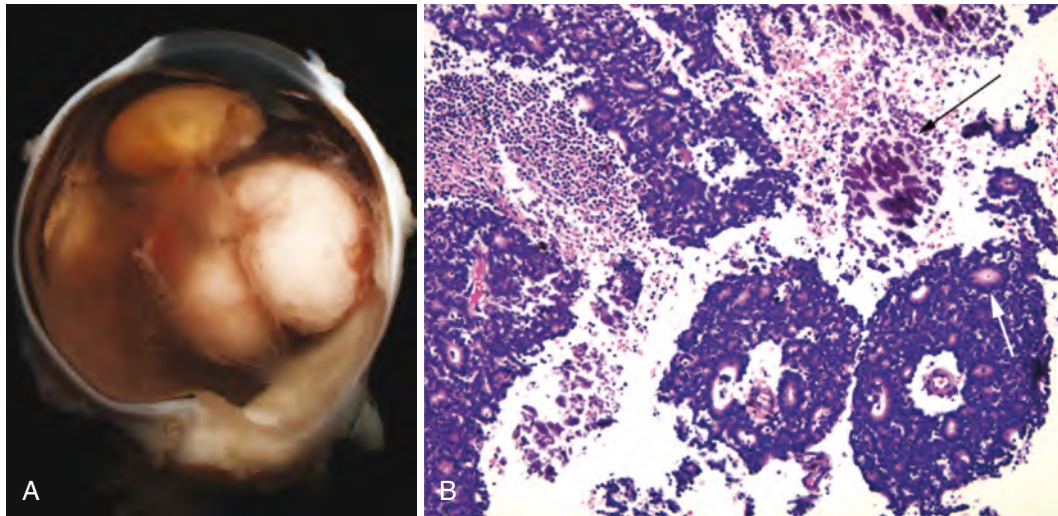


Fig. 29.26 Retinoblastoma. (A) Gross photograph of retinoblastoma. (B) Tumor cells are viable near blood vessels and become necrotic as the distance to the vessel increases. Dystrophic calcification (*black arrow*) is present in the zones of tumor necrosis. Flexner-Wintersteiner rosette, arrangements of a single layer of tumor cells around an apparent “lumen,” are seen throughout the tumor; the *white arrow* points to one such rosette.

- The appearance of the retina by ophthalmoscopy can be linked to specific pathologic changes: the change in caliber and color of retinal blood vessels may reflect varying degrees of arteriolosclerosis, and the location of hemorrhages and exudates in the retina is related to their sites within the retinal layers.
- Several significant causes of blindness result from pathologic intraocular angiogenesis, including proliferative diabetic retinopathy and exudative (wet) age-related macular degeneration, among many other conditions. VEGF antagonists may prevent visual loss in many of these conditions.
- Retinoblastoma is the most common primary intraocular tumor in children.
- Primary retinal lymphoma is an aggressive tumor that often involves the brain.

OPTIC NERVE

As a sensory tract of the central nervous system, the optic nerve is surrounded by meninges, and cerebrospinal fluid circulates around the nerve. The pathology of the optic nerve is similar to the pathology of the brain. For example, the most common primary neoplasms of the optic nerve are glioma (typically pilocytic astrocytomas) and meningioma.

Anterior Ischemic Optic Neuropathy

There are striking similarities between stroke and a condition known in ophthalmic terminology as *anterior ischemic optic neuropathy* (AION). As used clinically, the term AION includes a spectrum of injuries to the optic nerve varying from ischemia to infarction, similar to stroke. Thus, transient partial interruptions in blood flow to the optic nerve can produce episodes of transient vision loss, whereas a complete interruption of in blood flow can give rise to an optic nerve infarct. Zones of relative ischemia may surround segmental infarcts of the optic nerve. Optic nerve function in these poorly perfused but viable zones may recover. The optic nerve does not regenerate, and visual loss from infarction is permanent.

Interruption in the blood supply to the optic nerve can result from inflammation of the vessels that supply the optic nerve or from embolic or thrombotic events. Bilateral total infarcts of the optic nerve resulting in total blindness have been reported in temporal

arteritis, adding urgency to treating this condition with high doses of corticosteroids.

Papilledema

Edema of the head of the optic nerve may develop as a consequence of compression of the nerve (as when a primary neoplasm of the optic nerve produces unilateral disc edema) or from elevations of cerebrospinal fluid pressure surrounding the nerve (typically resulting in bilateral disc edema). The concentric increase in pressure encircling the nerve contributes to venous stasis and interferes with axoplasmic transport, leading to nerve head swelling. Typically, acute papilledema from increased intracranial pressure is not associated with visual loss. Ophthalmoscopically, the optic nerve head is swollen and hyperemic; by contrast, the optic nerve head in acute phases of anterior ischemic optic neuropathy appears swollen and pale because of decreased nerve perfusion (Fig. 29.27).

Glaucomatous Optic Nerve Damage

Glaucomatous optic nerve damage mostly stems from the elevated intraocular pressure but in a small group visual field and optic nerve changes typical of glaucoma develop with normal intraocular pressure. They are said to have normal-tension glaucoma. Conversely, some individuals with elevated intraocular pressure who are followed over long periods never develop visual field changes or optic nerve cupping. Therefore, it is clear that there is a spectrum of neuronal susceptibility to the effects of elevated intraocular pressure. Considerable research is now directed toward understanding mechanisms by which the optic nerve axons may be protected from injury.

MORPHOLOGY

Characteristically, there is a diffuse loss of ganglion cells and thinning of the retinal nerve fiber layer (Fig. 29.28), which can be measured by optical coherence tomography. The optic nerve is cupped and atrophic in advanced cases, a combination unique to glaucoma. Elevated intraocular pressure in infants and children can lead to diffuse enlargement of the eye (**buphthalmos**) or cornea (**megalocornea**). After the eye reaches its adult size, prolonged intraocular pressure elevation can lead to focal thinning of the sclera, and uveal tissue may line the ectatic sclera (**staphyloma**).

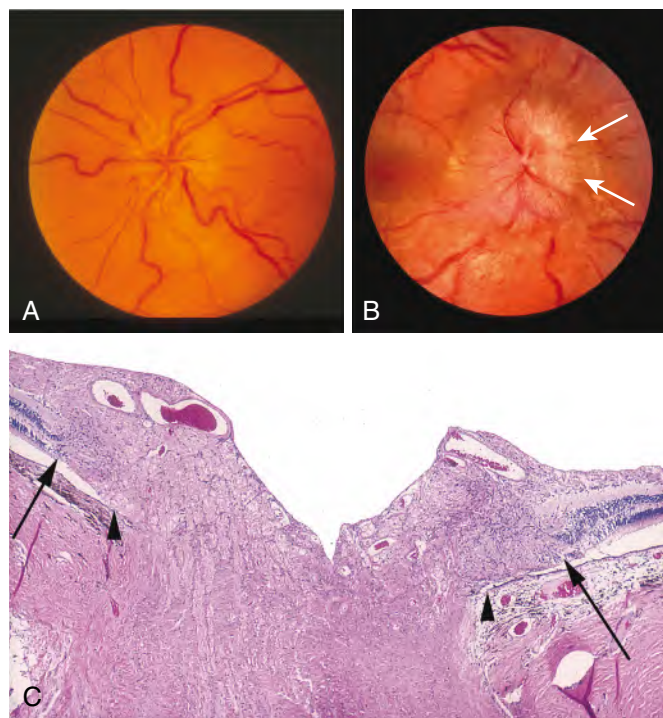


Fig. 29.27 The optic nerve in anterior ischemic optic neuropathy (AION) and papilledema. (A) In the acute phases of AION, the optic nerve may be swollen, but it is relatively pale because of decreased perfusion. (B) The optic nerve is typically swollen and hyperemic in papilledema, secondary to increased intracranial pressure. (C) The termination of Bruch membrane (*arrowheads*) is aligned with the beginning of the neurosensory retina (*arrows*). Here, the presence of swelling of the optic nerve due to papilledema has displaced the retina laterally. This displacement is the explanation for the blurred optic nerve margins that are seen in this condition. (A–B, Courtesy Dr. Sohan S. Hayreh, Department of Ophthalmology and Visual Science, University of Iowa, Iowa City. C, From the Armed Forces Institute of Pathology teaching collection.)

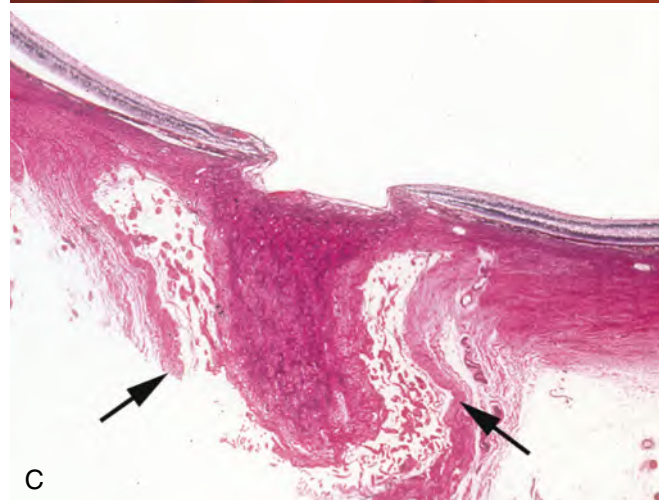
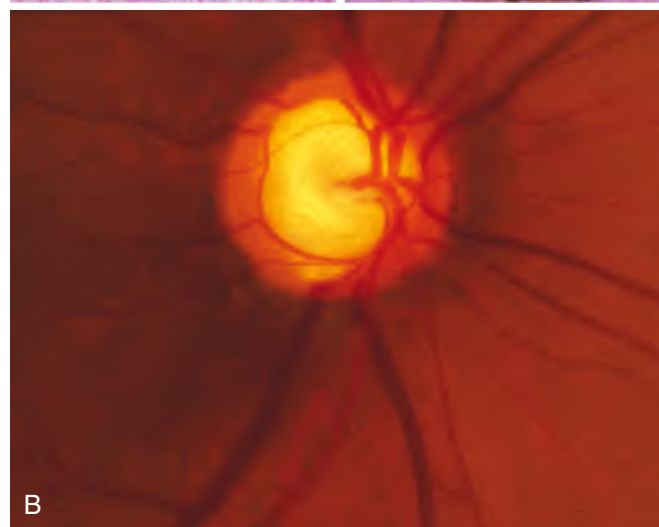
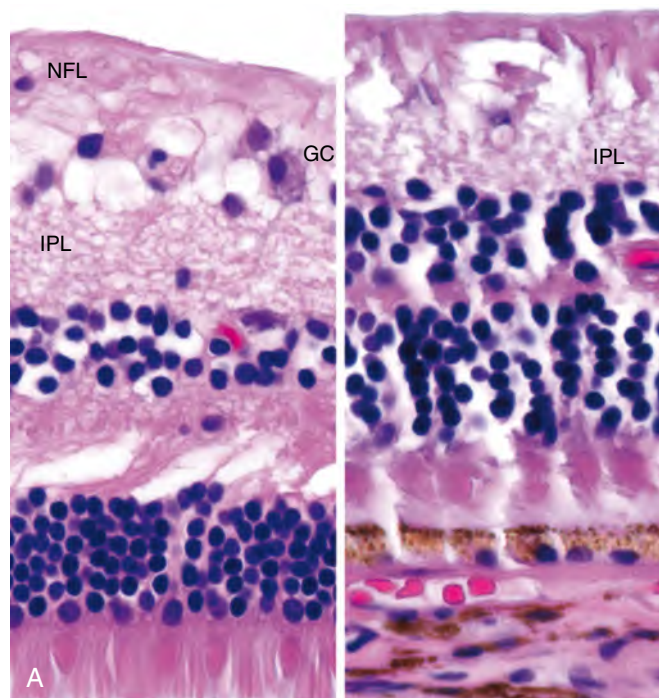


Fig. 29.28 The retina and optic nerve in glaucoma. (A) *Left*, normal retina; *right*, the retina in long-standing glaucoma (same magnification). The glaucomatous retina is thinned and areas corresponding to the nerve fiber layer (NFL) and ganglion cell layer (GC) are atrophic; the inner plexiform layer (IPL) is labeled for reference. (B) Glaucomatous optic nerve has cupping which results in part from the loss of retinal ganglion cells, the axons of which populate the optic nerve. (C) The *arrows* point to the dura of the optic nerve. Notice the wide subdural space resulting from atrophy of the nerve. There is a striking degree of cupping on the surface of the nerve as a consequence of long-standing glaucoma.

Other Optic Neuropathies

Optic neuropathy refers to optic nerve damage that may be inherited or secondary to nutritional deficiencies or toxins such as methanol. Individuals may have severe visual compromise. Central visual acuity is lost if the nerve fibers originating from the macula are affected.

Leber hereditary optic neuropathy results from inherited mitochondrial gene mutations (Chapters 5 and 28). Since neuronal health depends oxidative phosphorylation carried out by mitochondria, neurons are particularly susceptible mitochondrial disorders, including optic neuropathy. Leber optic neuropathy shows a maternal inheritance pattern typical of mitochondrial gene mutations. Males are affected far more commonly than females. The usual age of onset is between 10 and 30 years. It begins with clouding of vision that may progress to total vision loss.

Optic Neuritis

Many unrelated conditions have historically been grouped under the heading of optic neuritis. Unfortunately, the term suggests optic nerve inflammation, which might not accurately describe the pathophysiologic changes. In everyday clinical usage, optic *neuritis* describes a loss of vision secondary to demyelination of the optic nerve. Multiple sclerosis is one of the most important causes of optic neuritis (Chapter 28). Indeed, optic neuritis may be the first manifestation of this disease. The 10-year risk of developing multiple sclerosis after the first attack of optic neuritis increases if the affected person has concomitant evidence of brain lesions as detected by magnetic resonance imaging. Individuals with a single episode of optic nerve demyelination may recover vision and remain disease-free.

KEY CONCEPTS

Optic Nerve

- Anterior ischemic optic neuropathy (AION) refers to a spectrum of ischemic injuries to the optic nerve varying from transient ischemia to infarction.
- Bilateral swelling of the optic nerve head, known as papilledema, may develop due to elevated cerebrospinal fluid pressure and stasis of axoplasmic transport within the optic nerve. Unilateral optic nerve head swelling may result from optic nerve compression, such as in primary optic nerve tumors.
- In chronic glaucoma, the optic nerve may atrophy, and the cup on the surface of the nerve may enlarge and deepen.
- Optic neuropathy may be inherited (as in Leber hereditary optic neuropathy) or result from nutritional deficiencies or toxins such as methanol.

THE END-STAGE EYE: PHTHISIS BULBI

Trauma, intraocular inflammation, chronic retinal detachment, and many other conditions can give rise to an internally disorganized, small atrophic eye called phthisis bulbi. Congenitally small eyes (i.e., hypoplastic or microphthalmic eyes) are generally not disorganized internally. Phthisical eyes typically show the following changes: the presence of exudate or blood between the ciliary body and sclera and the choroid and sclera (*ciliochoroidal effusion*); the presence of a membrane extending across the eye from one aspect of the ciliary body to the other (*cyclitic membrane*); chronic retinal detachment; optic nerve atrophy; the presence of intraocular bone, which is thought to originate from osseous metaplasia of the RPE; and a thickened sclera, especially posteriorly. Ciliochoroidal effusion is typically associated with the physiologic state of low intraocular pressure (*hypotony*). The standard pull of the extraocular muscles on a hypotonic eye may render the appearance of the eye square rather than round.

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